Predicting College Student Performance Using Machine Learning Techniques

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University of the Incarnate Word

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Open University Learning Analytics Dataset

About Open University

- Headquarters based at Walton Hall in Milton Keynes, Buckinghamshire, England
- One of the largest universities in Europe with around 200,000 students enrolled
- Popular for its flexible distance learning programs

About Open University Learning Analytics Dataset

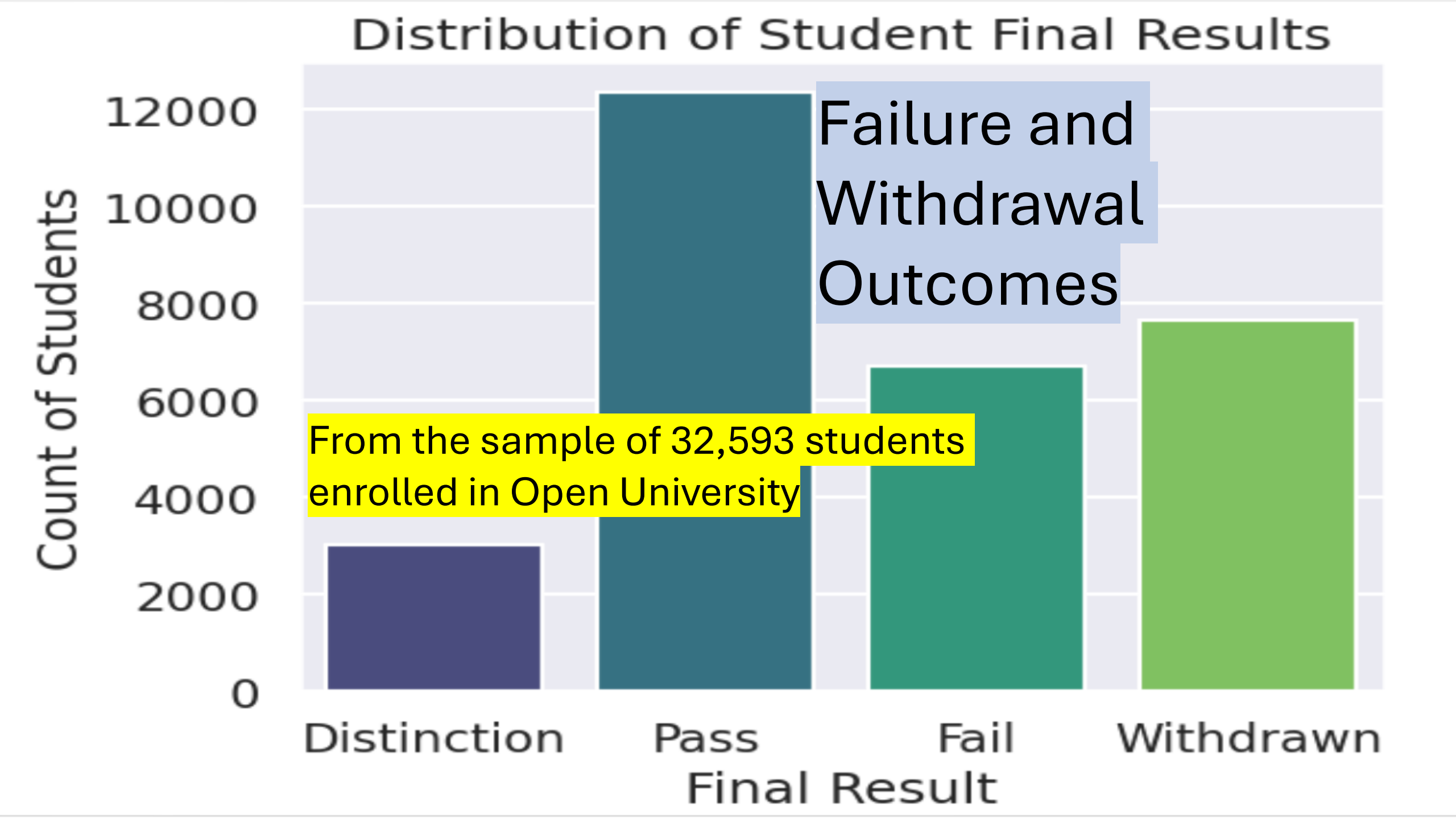
- Contains student demographic, registration, assessment and daily activities (interactions) with Open University's VLE (Virtual Learning Environment) courses from the years 2013 and 2014
- The data is anonymized
- The data contains 22 courses with 32,593 registered students

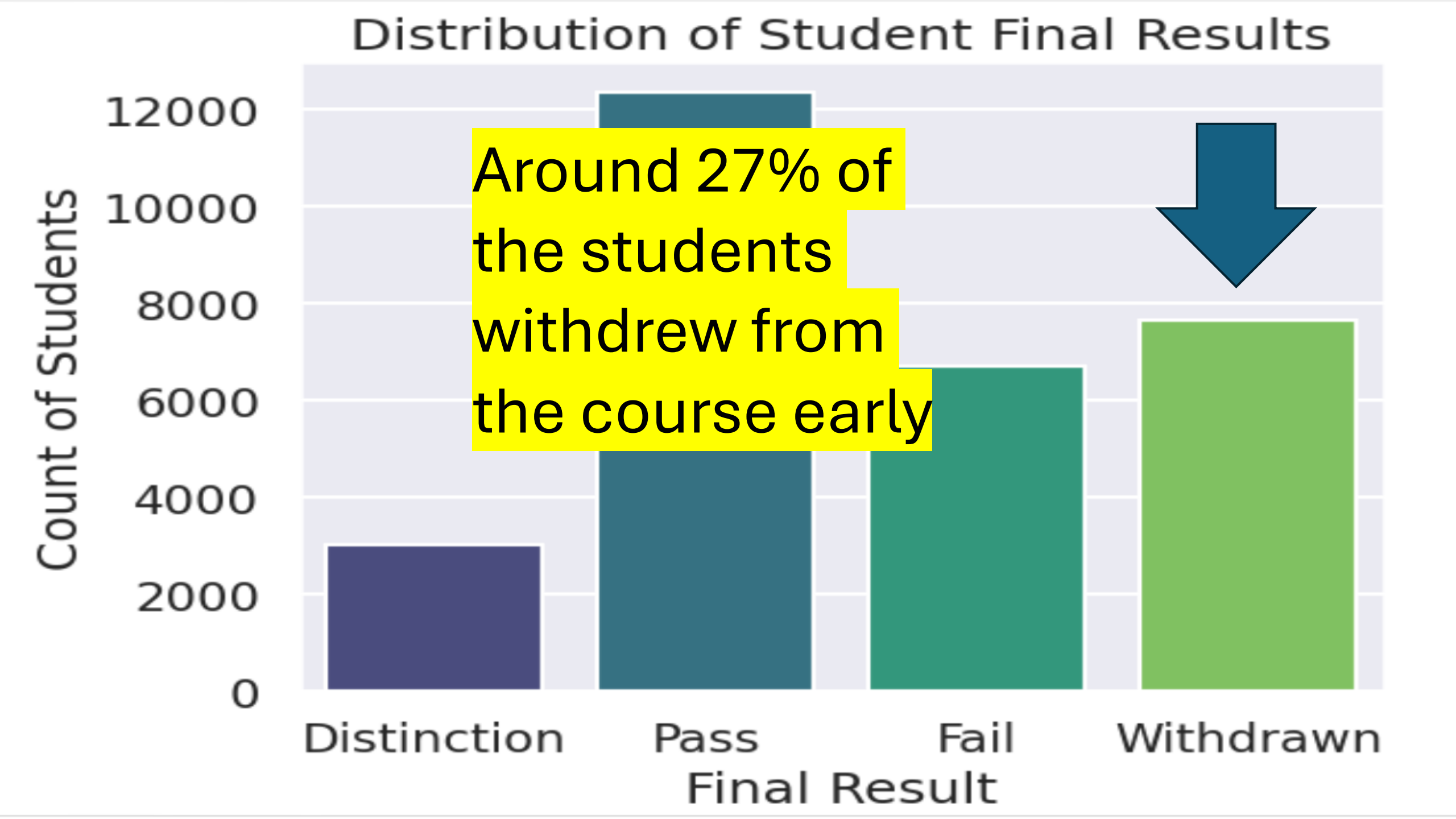
Exploratory Data Analysis

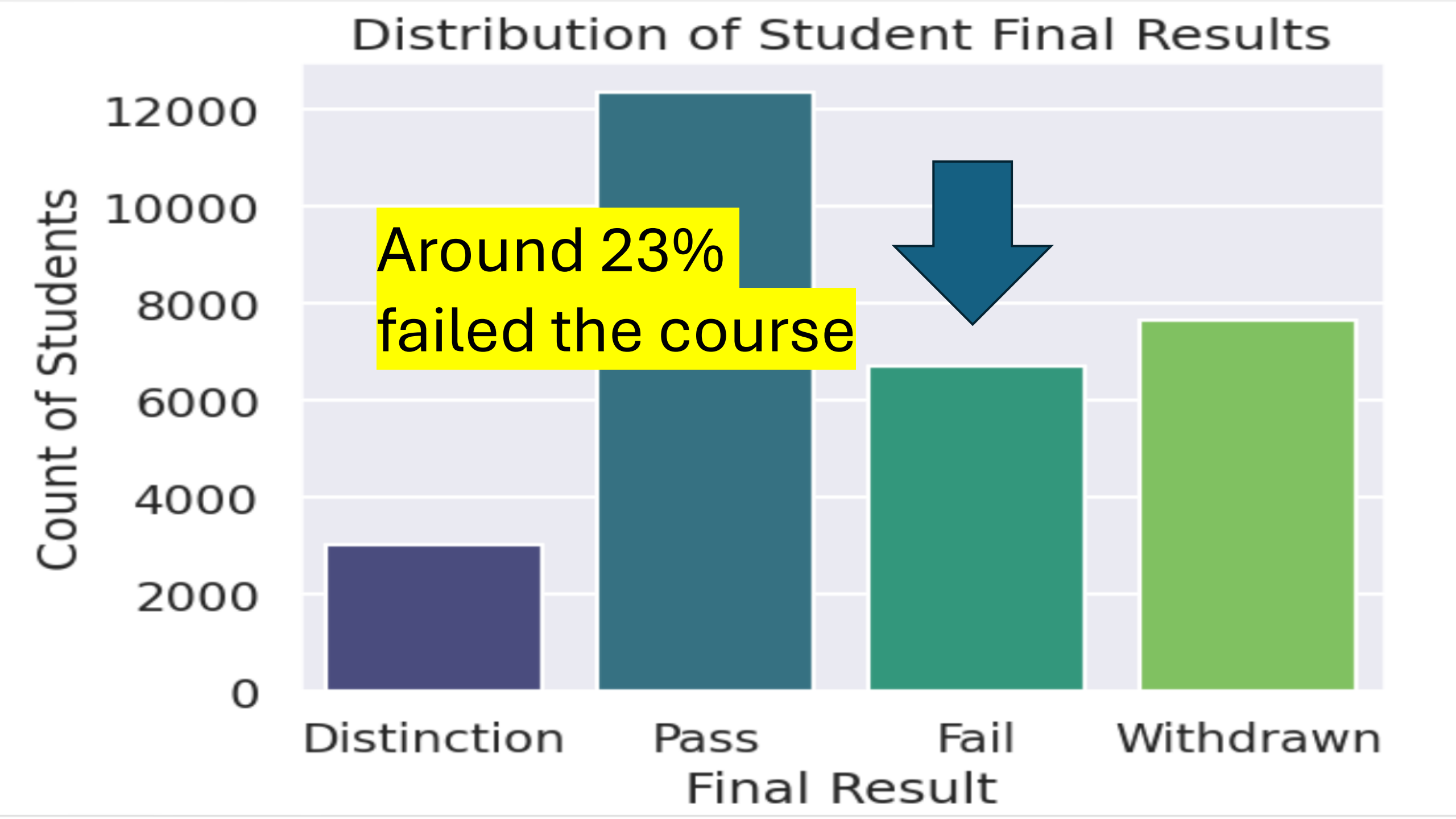
Objectives

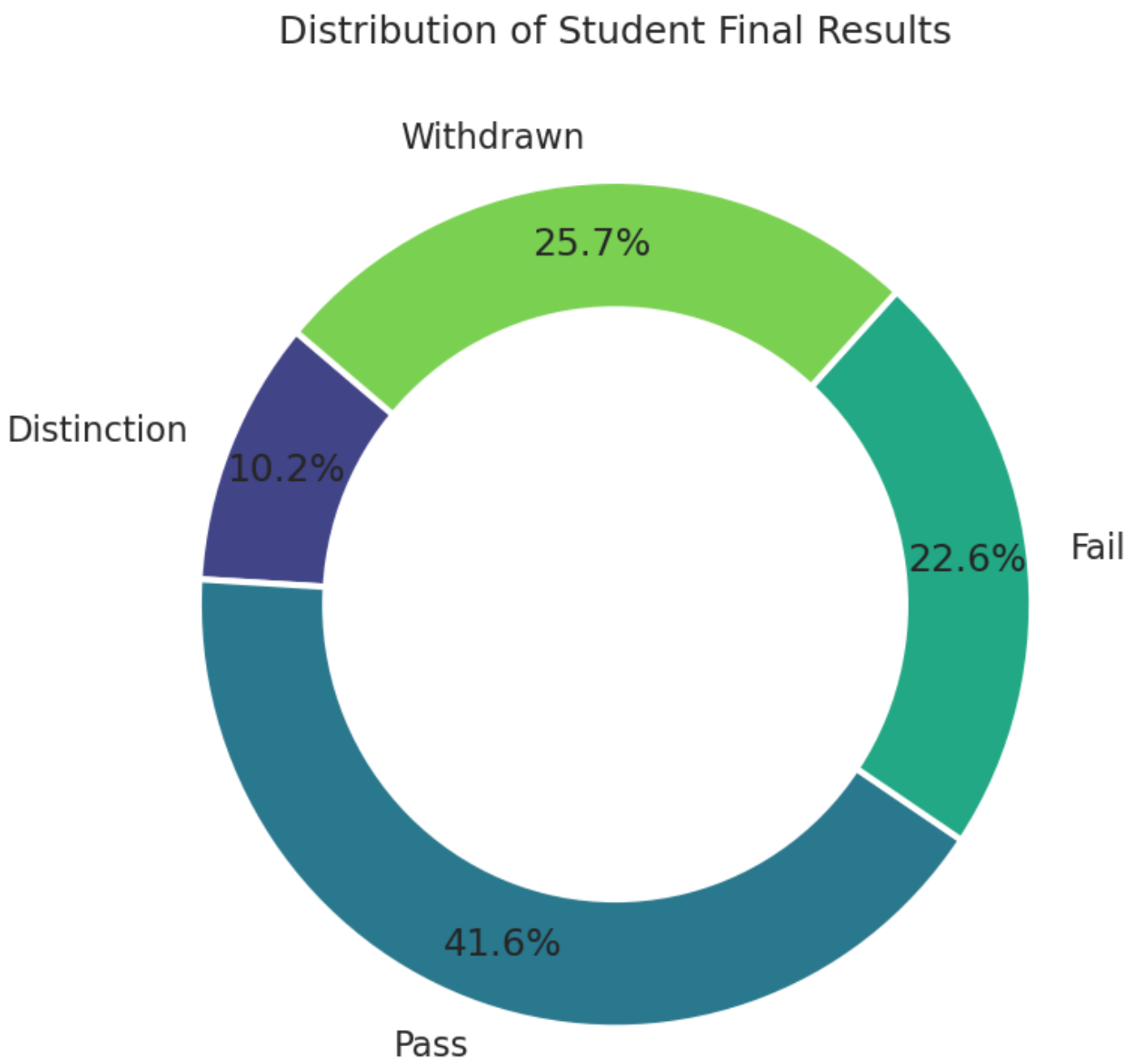
Using Exploratory Data Analysis, our goal is to:

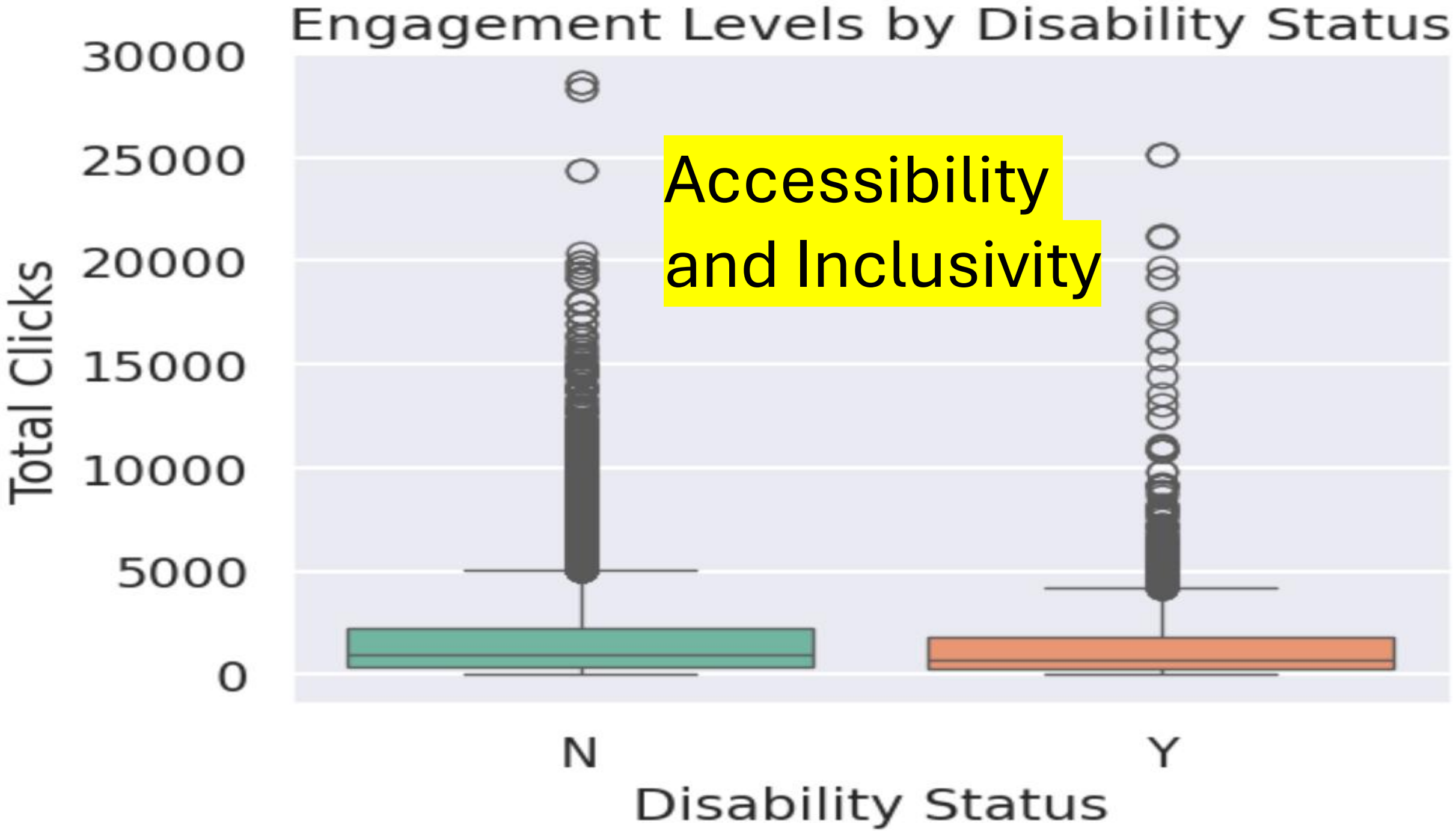
- Identify the course grade outcomes, including the withdrawal and failure rates.
- Determine if the course is accessible and inclusive, regardless of student background or disability.
- Identify study habits that lead to course success.
- Identify and recommend revisions to course navigation for increasing student success

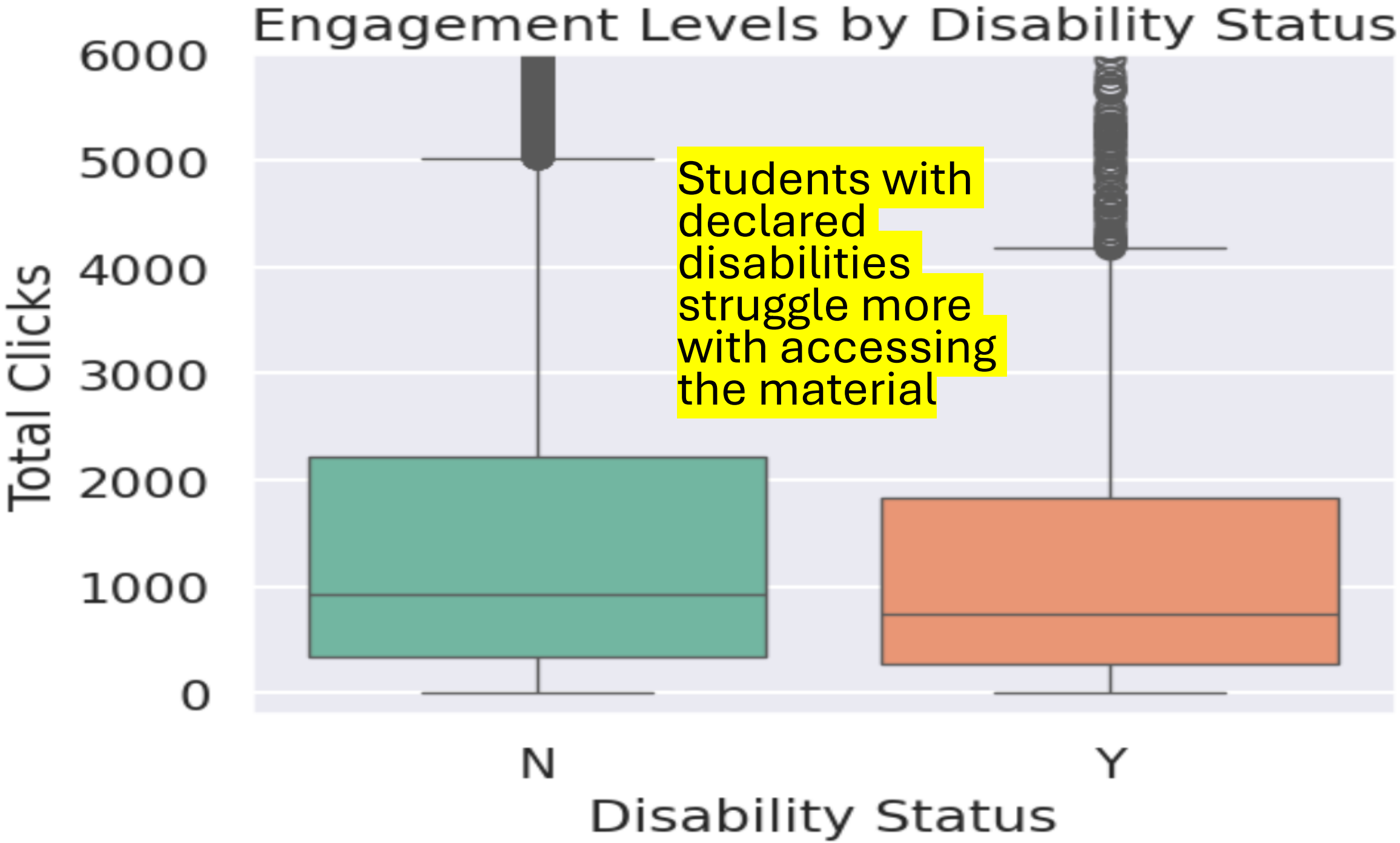

Distribution of Student Final Results
Failure and Withdrawal Outcomes
From the sample of 32,593 students enrolled in Open University
Count of Students
12000
10000
8000
6000
4000
2000
0
Distinction
Pass
Fail
Withdrawn
Final Result

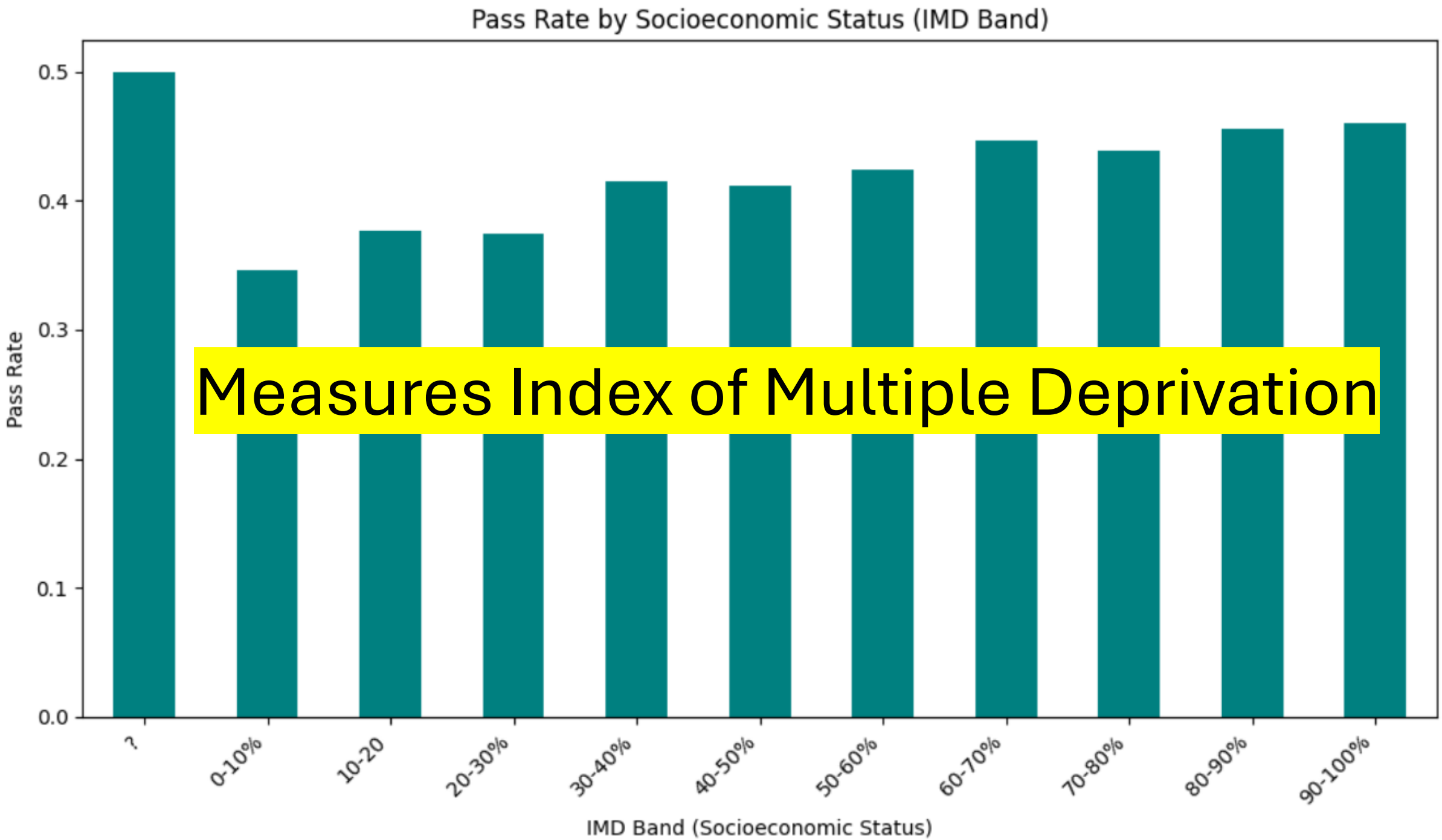

Distribution of Student Final Results
Count of Students
12000
10000
8000
6000
4000
2000
0
Distinction
Pass
Fail
Withdrawn
Final Result
Around 27% of the students withdrew from the course early


Distribution of Student Final Results
Count of Students
12000
10000
8000
6000
4000
2000
0
Around 23%
failed the course
Distinction
Pass
Fail
Withdrawn
Final Result

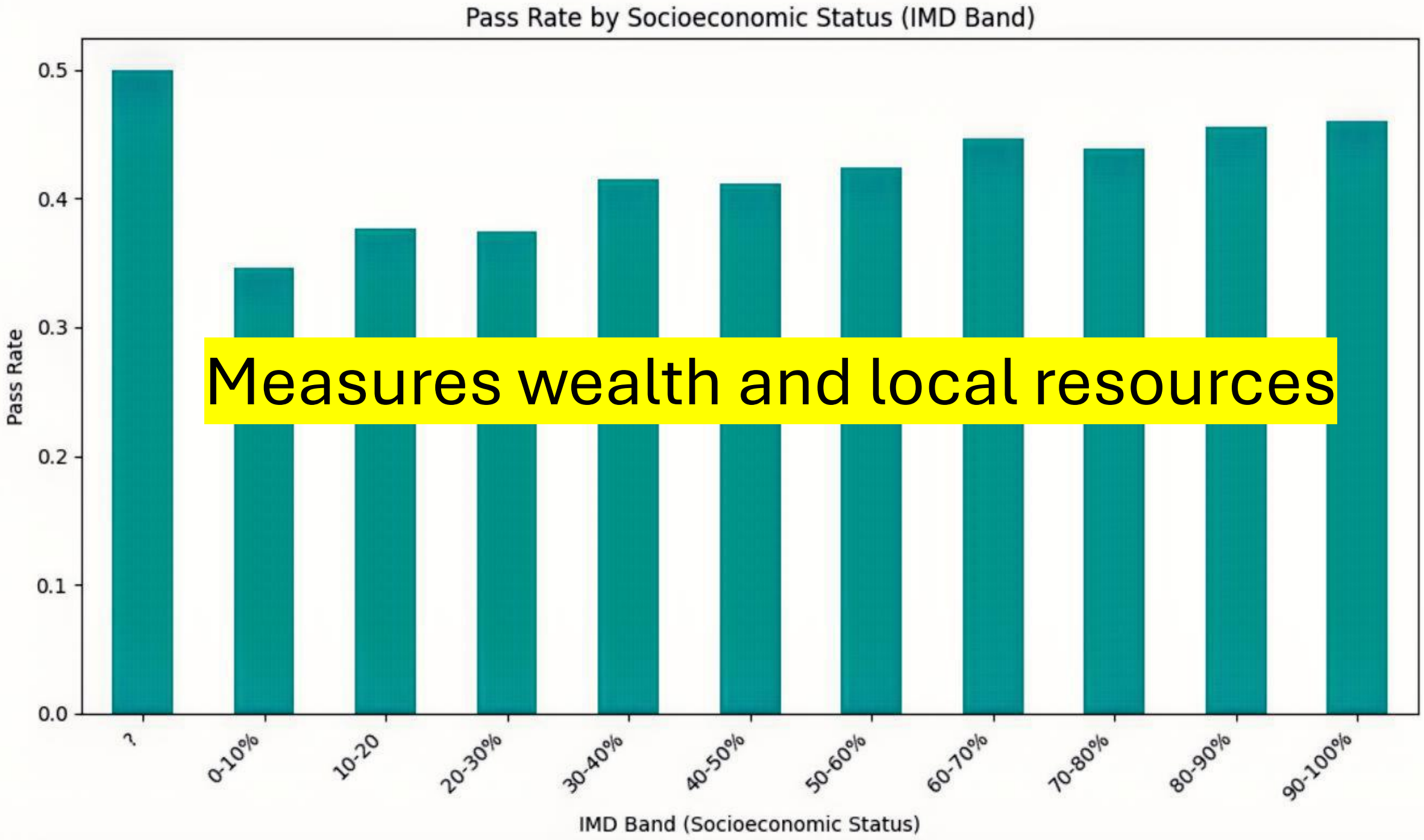

Distribution of Student Final Results
Withdrawn
25.7%
Distinction
10.2%
Fail
22.6%
41.6%
Pass


Engagement Levels by Disability Status
Accessibility
and Inclusivity
Total Clicks
30000
25000
20000
15000
10000
5000
0
N
Y
Disability Status

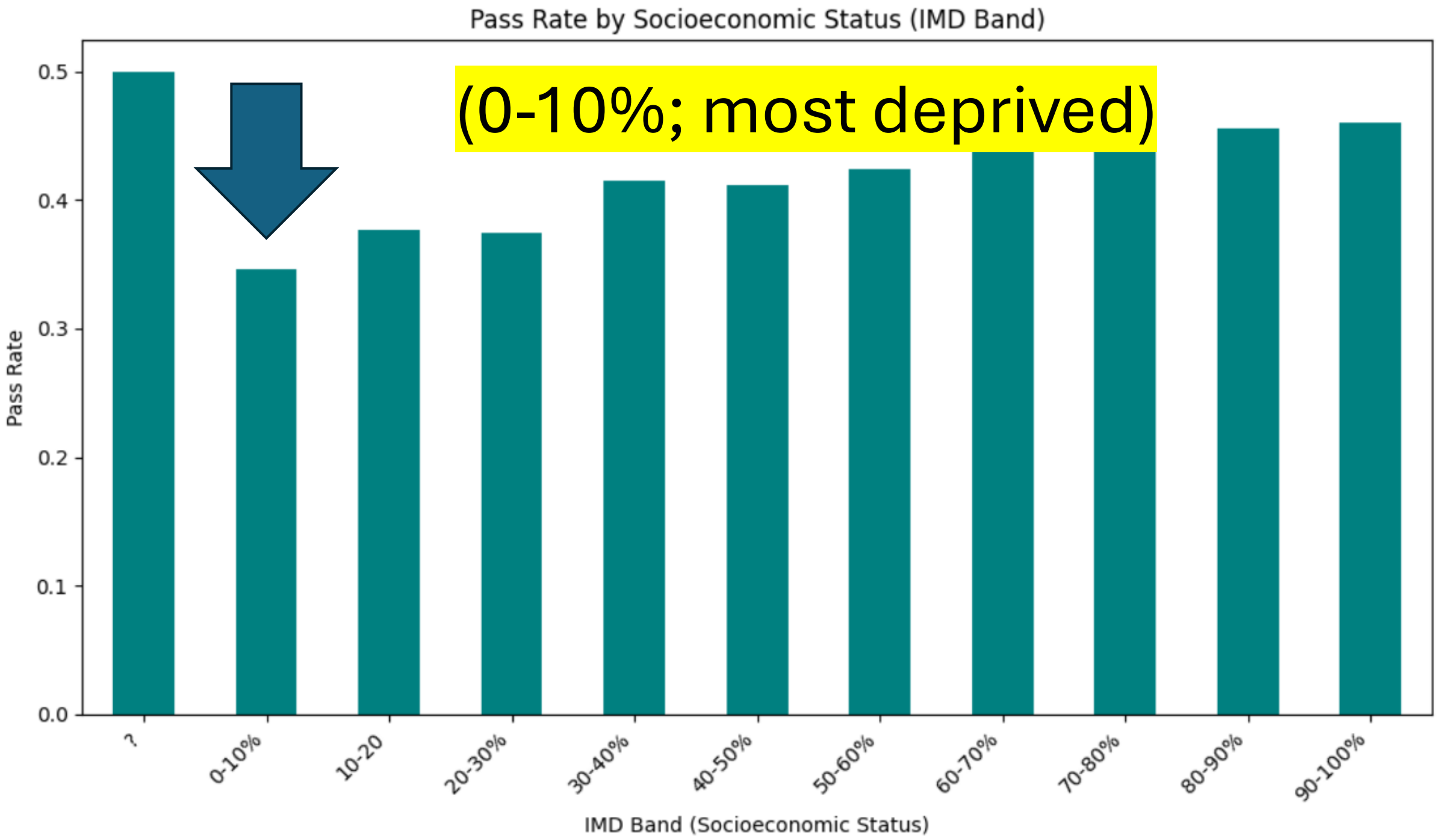

Engagement Levels by Disability Status
Total Clicks
6000
5000
4000
3000
2000
1000
0
N
Y
Disability Status
Students with declared disabilities struggle more with accessing the material

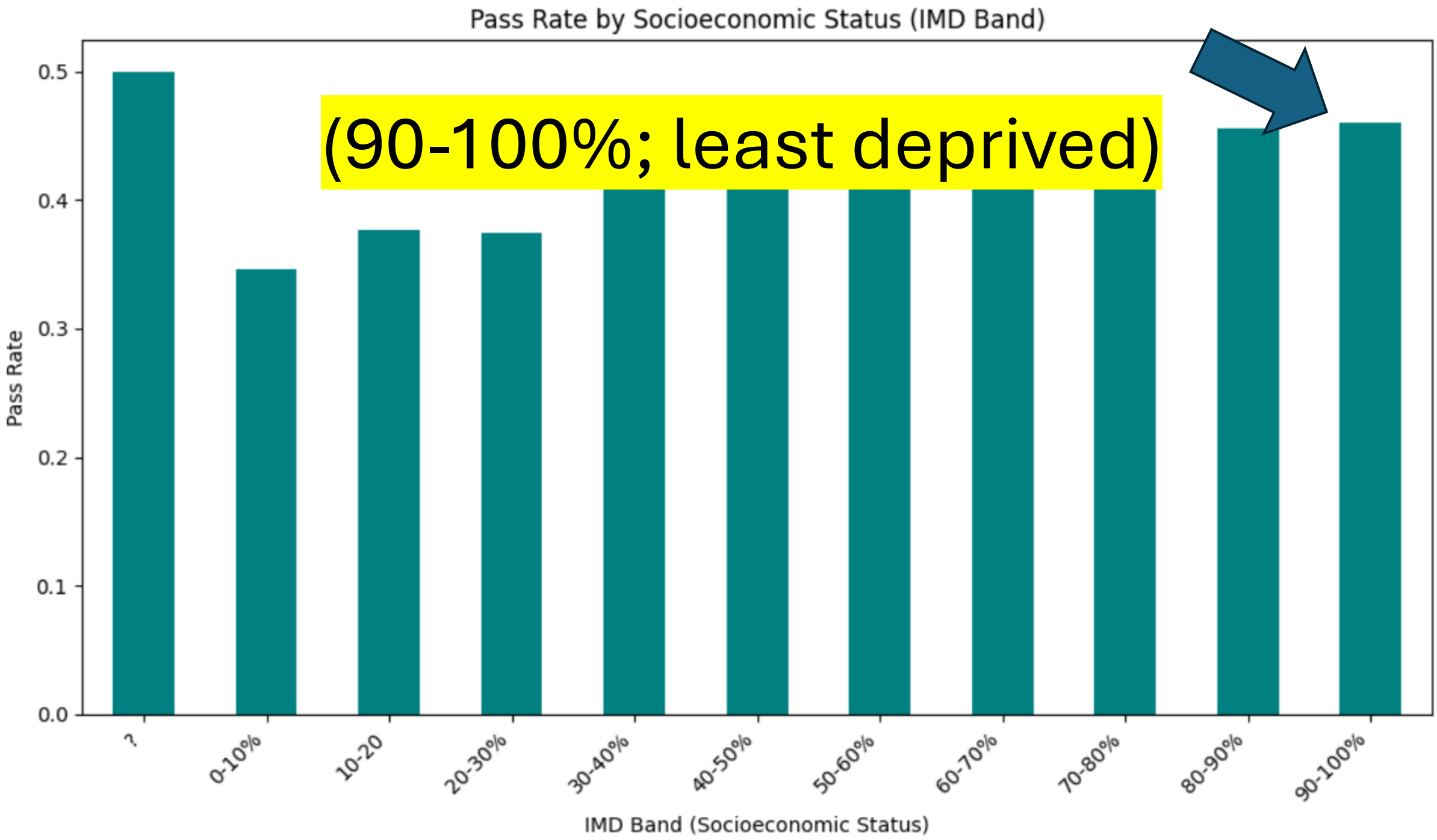


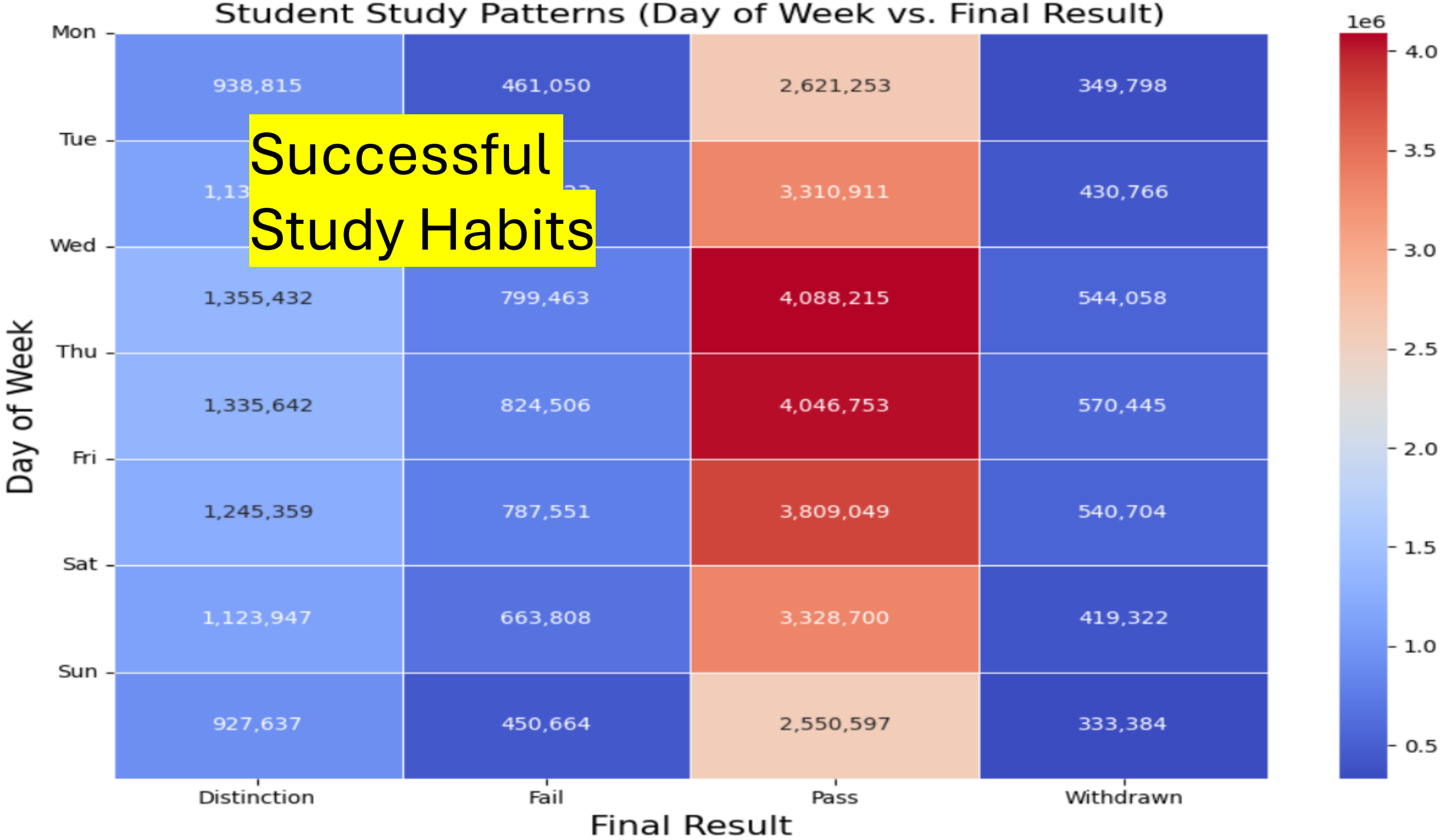
Measures Index of Multiple Deprivation

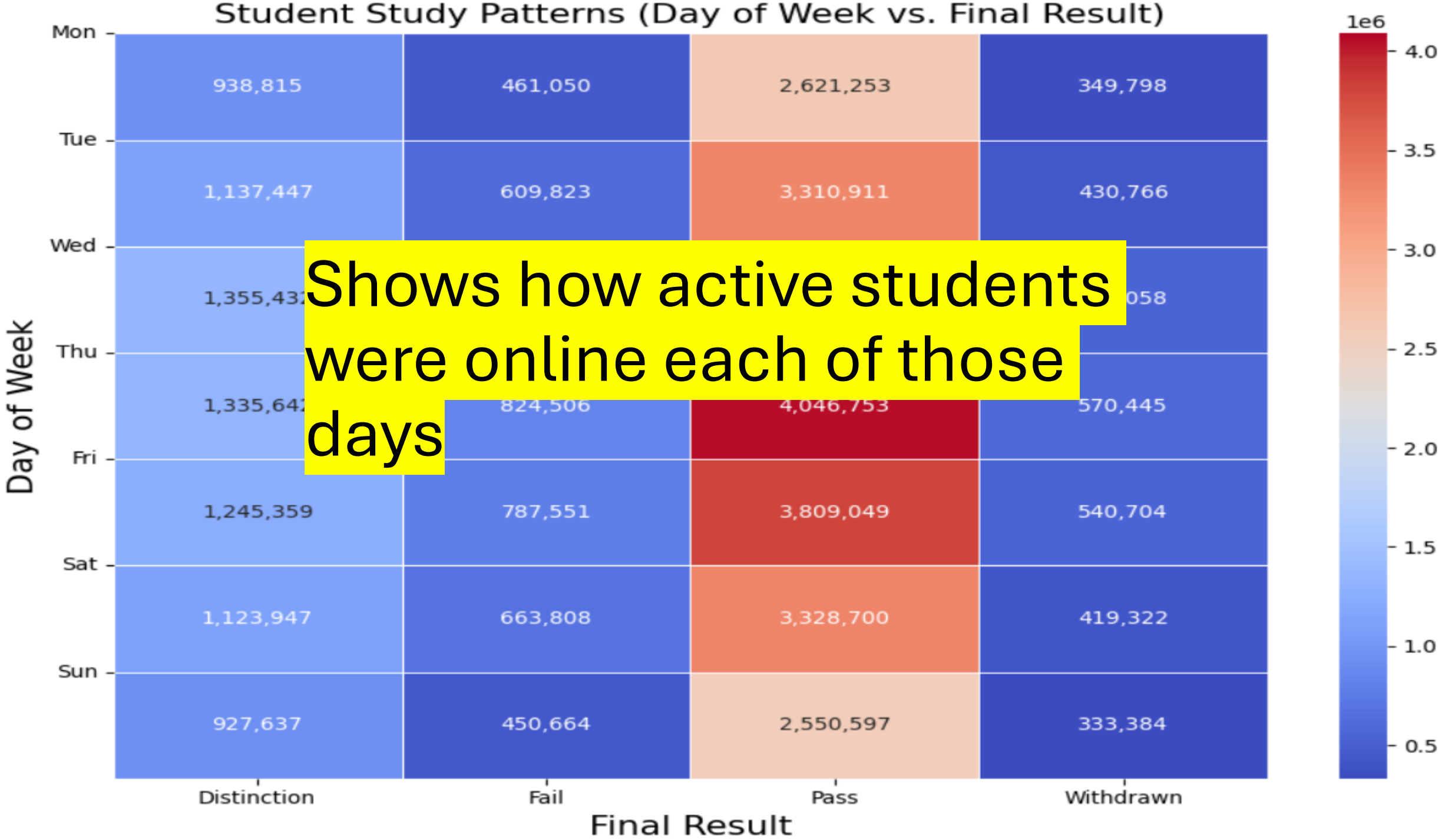


Measures wealth and local resources

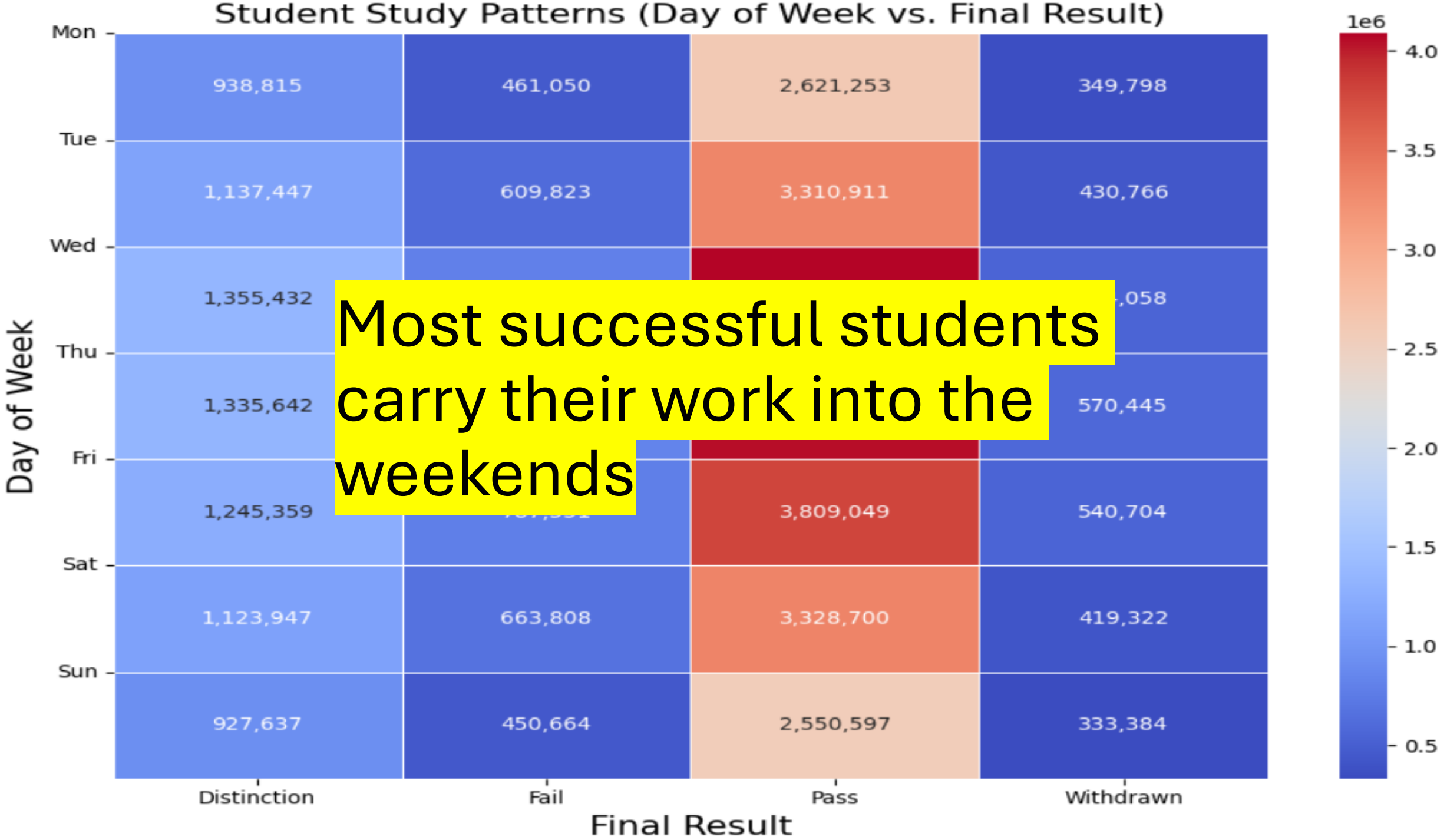

Pass Rate by Socioeconomic Status (IMD Band)
(0-10%; most deprived)
Pass Rate
0.0
0.1
0.2
0.3
0.4
0.5
?
0-10%
10-20
20-30%
30-40%
40-50%
50-60%
60-70%
70-80%
80-90%
90-100%
IMD Band (Socioeconomic Status)


Pass Rate by Socioeconomic Status (IMD Band)
(90-100%; least deprived)
Pass Rate
0.0
0.1
0.2
0.3
0.4
0.5
?
0-10%
10-20
20-30%
30-40%
40-50%
50-60%
60-70%
70-80%
80-90%
90-100%
IMD Band (Socioeconomic Status)

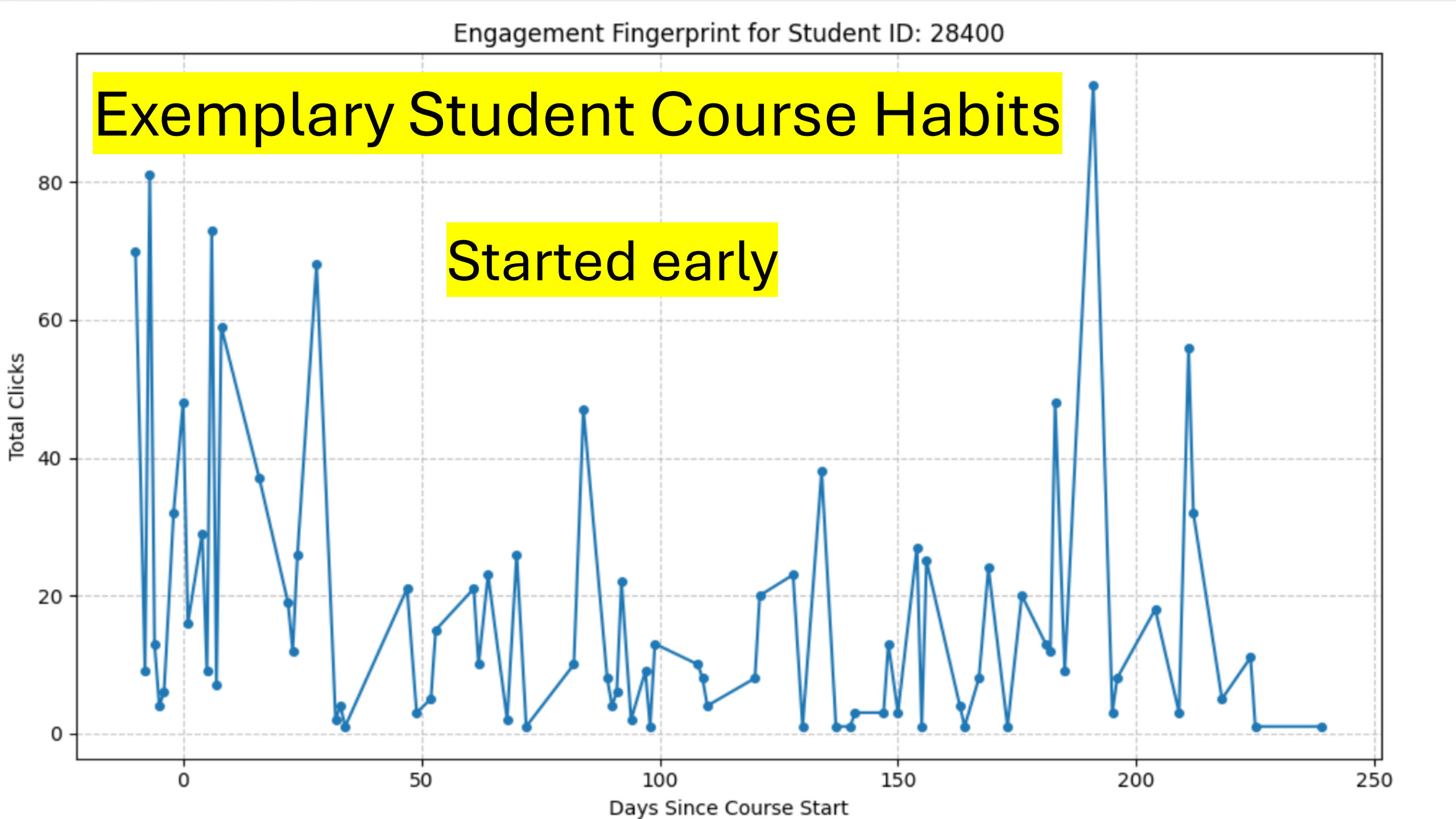

Student Study Patterns (Day of Week vs. Final Result)
Successful Study Habits
Day of Week
Mon
Tue
Wed
Thu
Fri
Sat
Sun
938,815
461,050
2,621,253
349,798
1,13
3,310,911
430,766
1,355,432
799,463
4,088,215
544,058
1,335,642
824,506
4,046,753
570,445
1,245,359
787,551
3,809,049
540,704
1,123,947
663,808
3,328,700
419,322
927,637
450,664
2,550,597
333,384
Distinction
Fail
Pass
Withdrawn
Final Result
1e6
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5

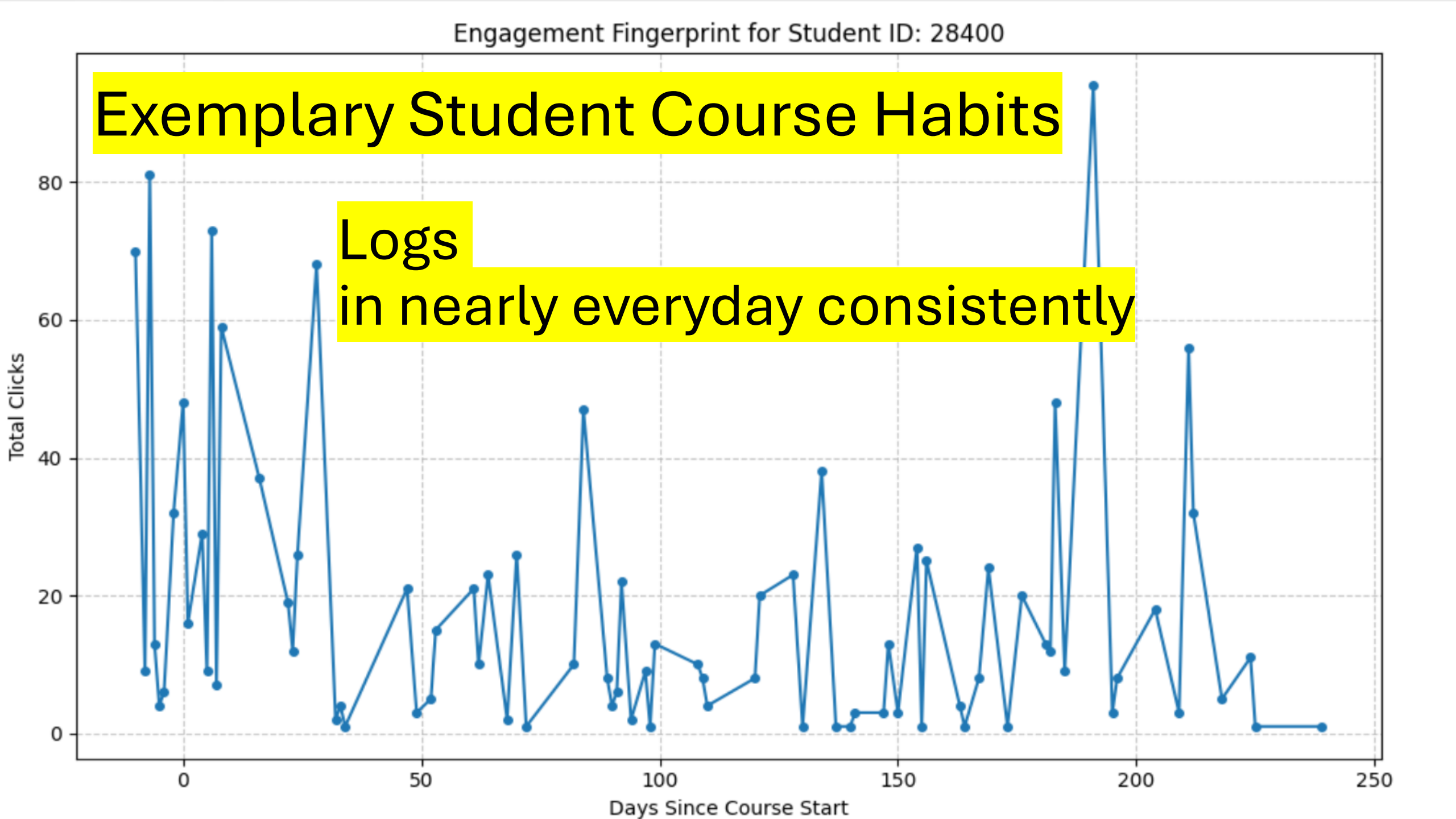


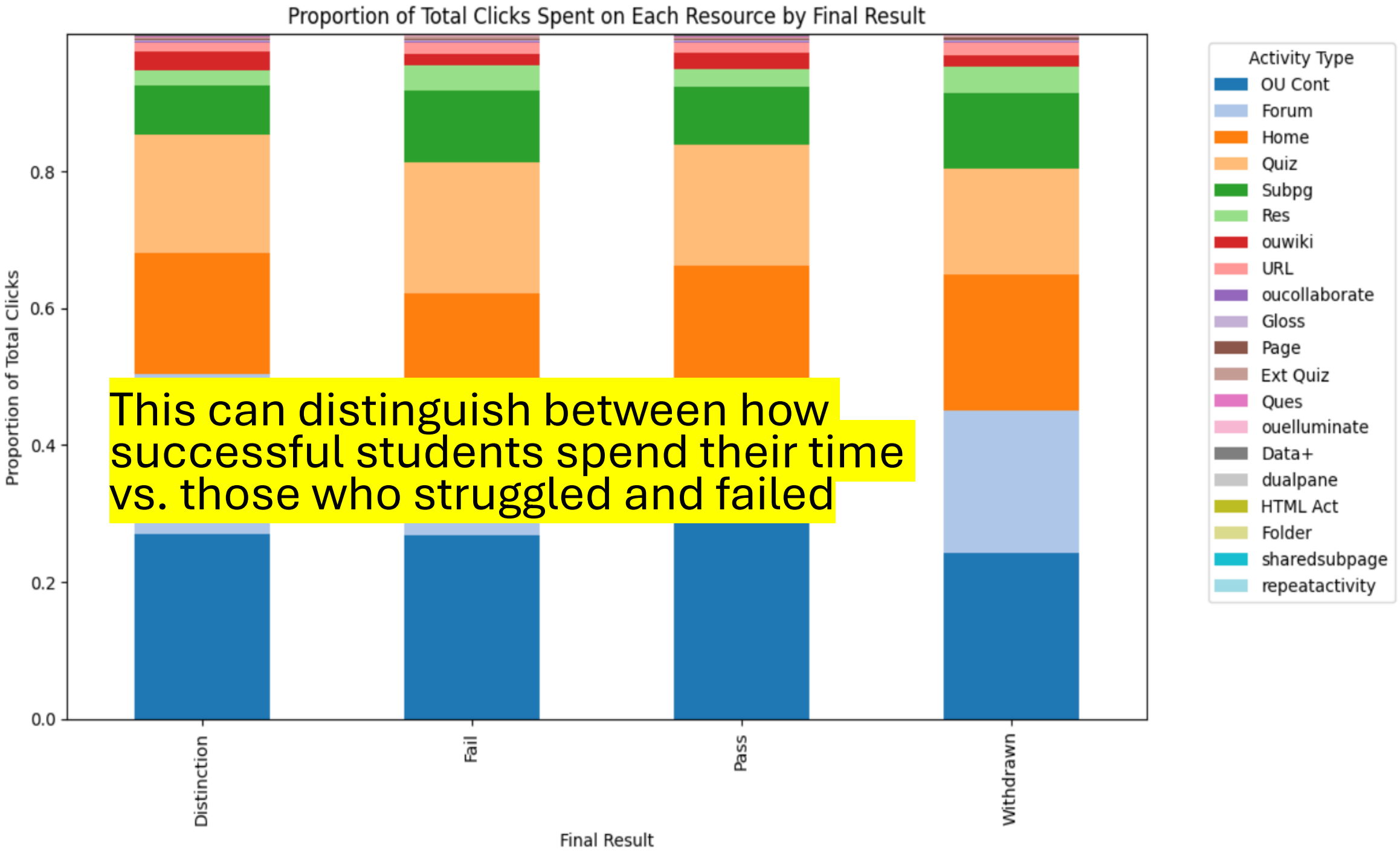
Shows how active students were online each of those days



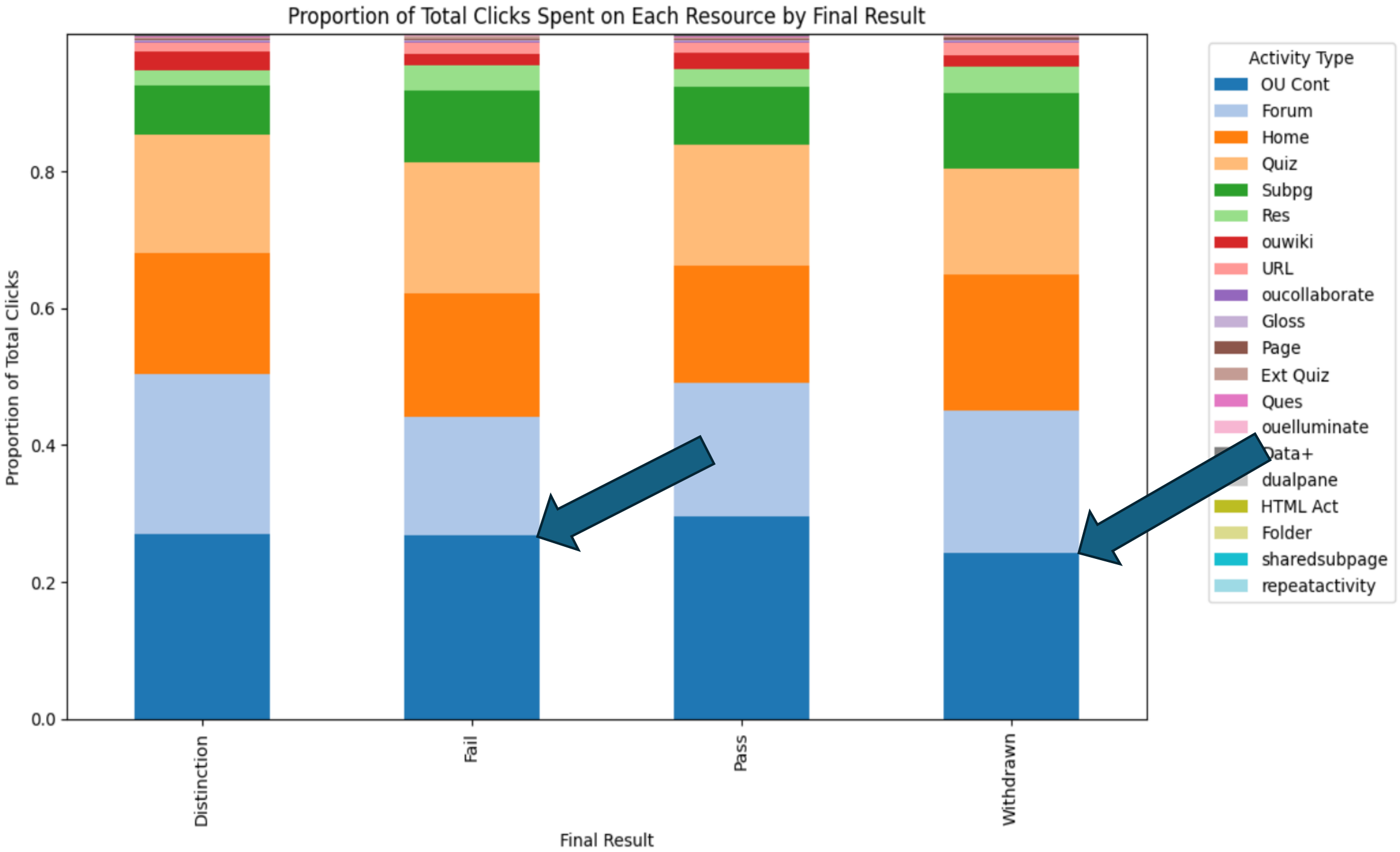
Most successful students carry their work into the weekends

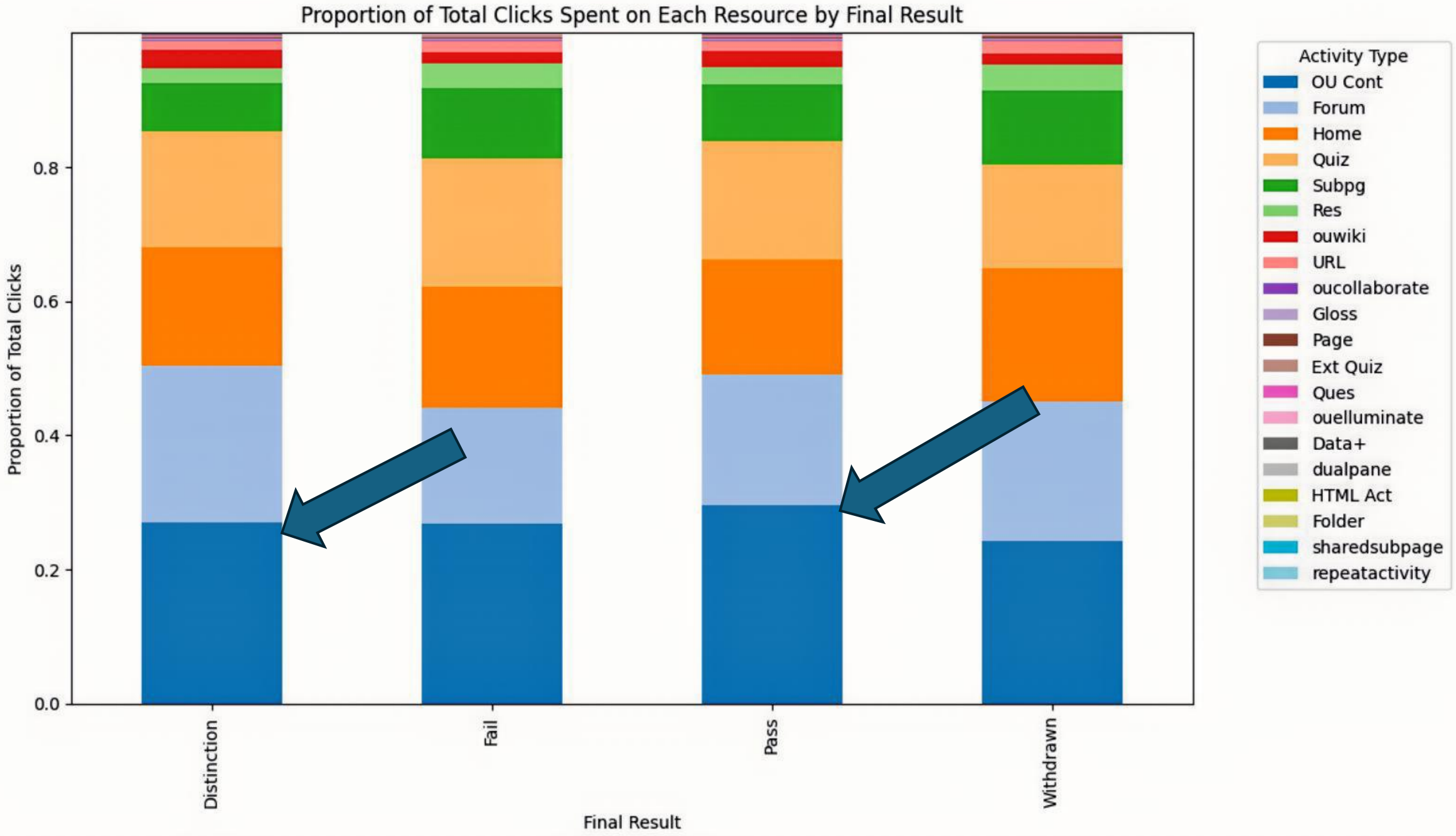

Engagement Fingerprint for Student ID: 28400
Exemplary Student Course Habits
Started early
Total Clicks
0
20
40
60
80
0
50
100
150
200
250
Days Since Course Start


Engagement Fingerprint for Student ID: 28400
Exemplary Student Course Habits
Logs
in nearly everyday consistently
Total Clicks
0
20
40
60
80
0
50
100
150
200
250
Days Since Course Start



This can distinguish between how successful students spend their time vs. those who struggled and failed


Proportion of Total Clicks Spent on Each Resource by Final Result
Proportion of Total Clicks
0.0
0.2
0.4
0.6
0.8
Distinction
Fail
Pass
Withdrawn
Final Result
Activity Type
OU Cont
Forum
Home
Quiz
Subpg
Res
ouwiki
URL
oucollaborate
Gloss
Page
Ext Quiz
Ques
ouelluminate
Data+
dualpane
HTML Act
Folder
sharedsubpage
repeatactivity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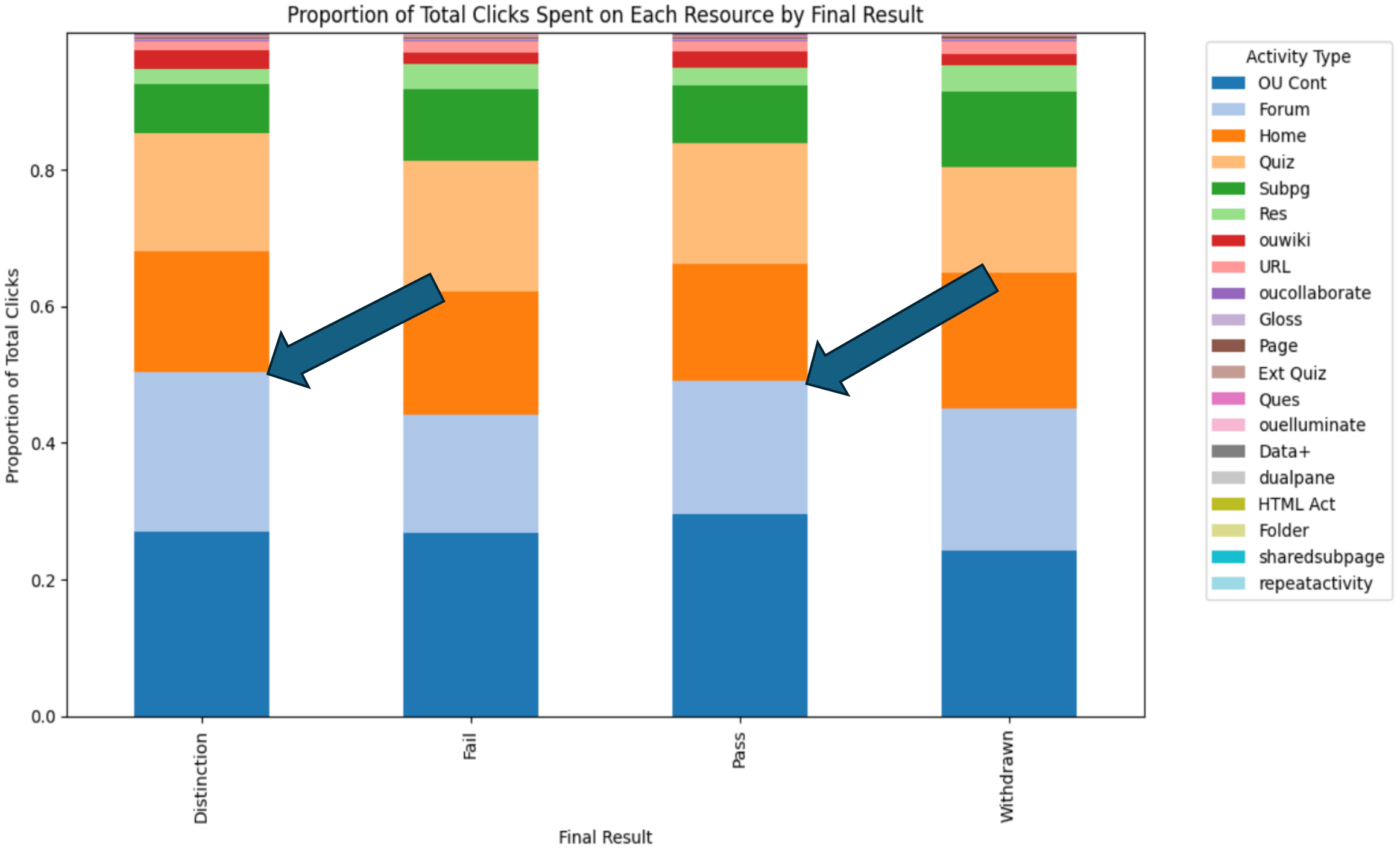
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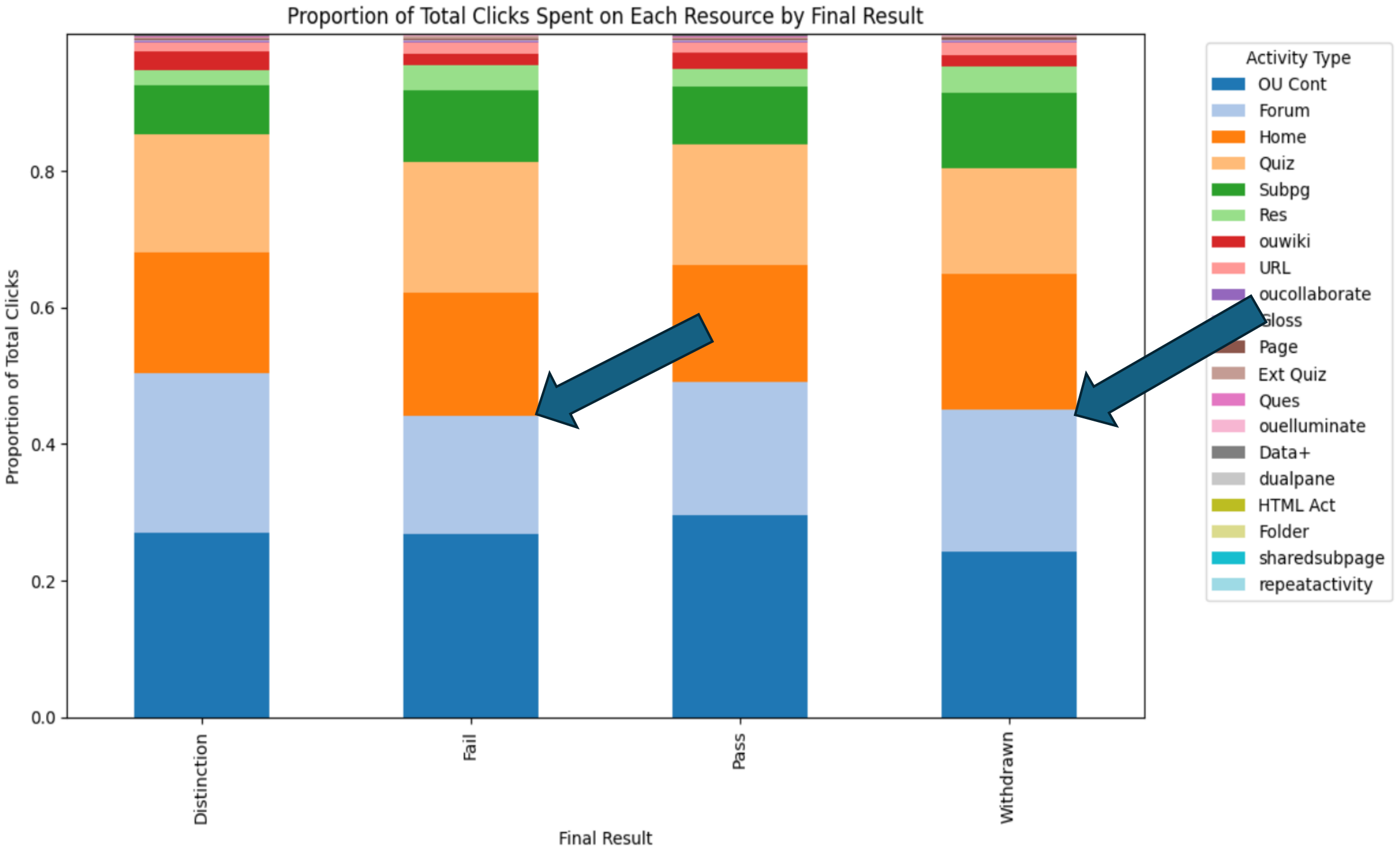
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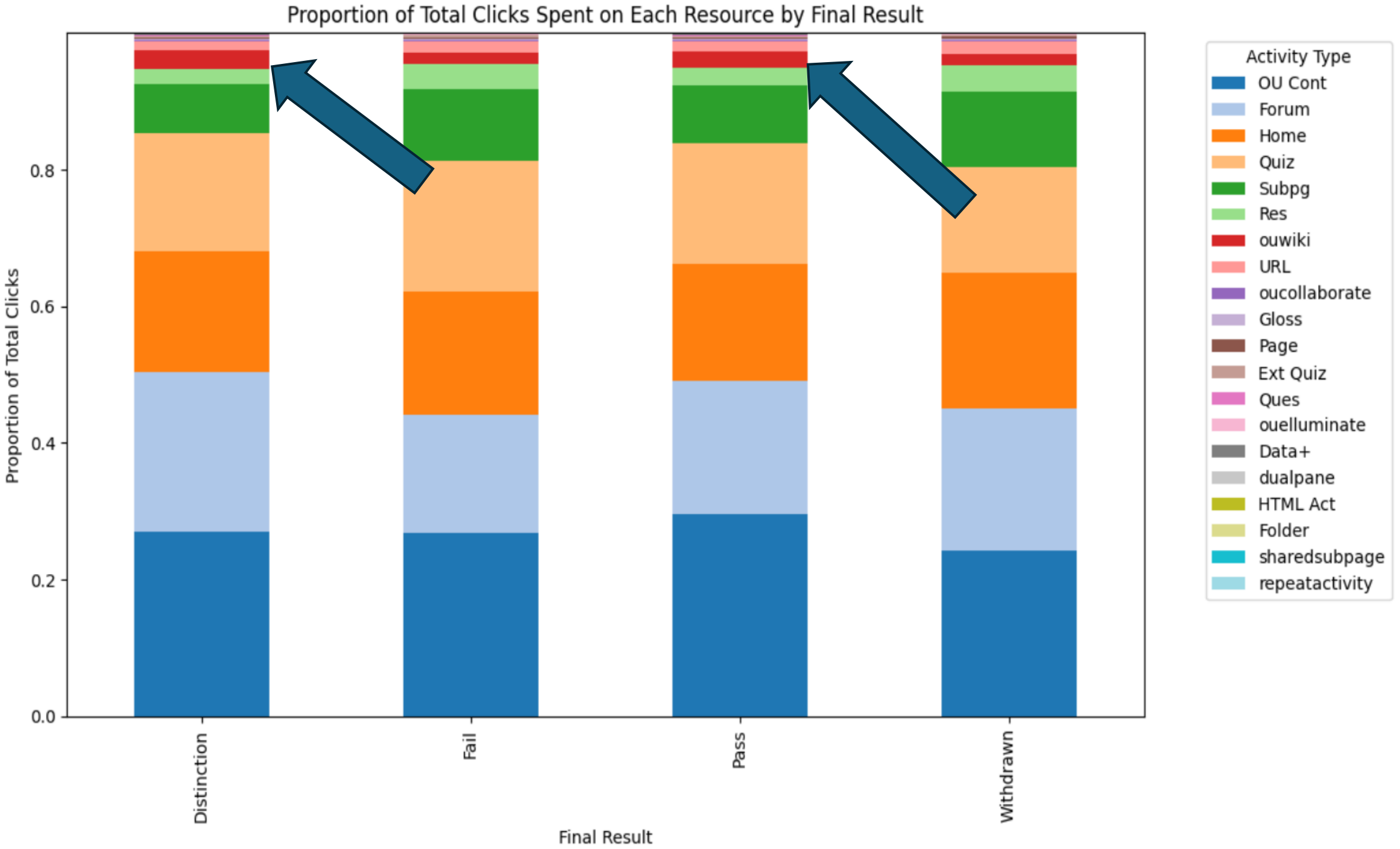
Proportion of Total Clicks Spent on Each Resource by Final Result
Proportion of Total Clicks
0.0
0.2
0.4
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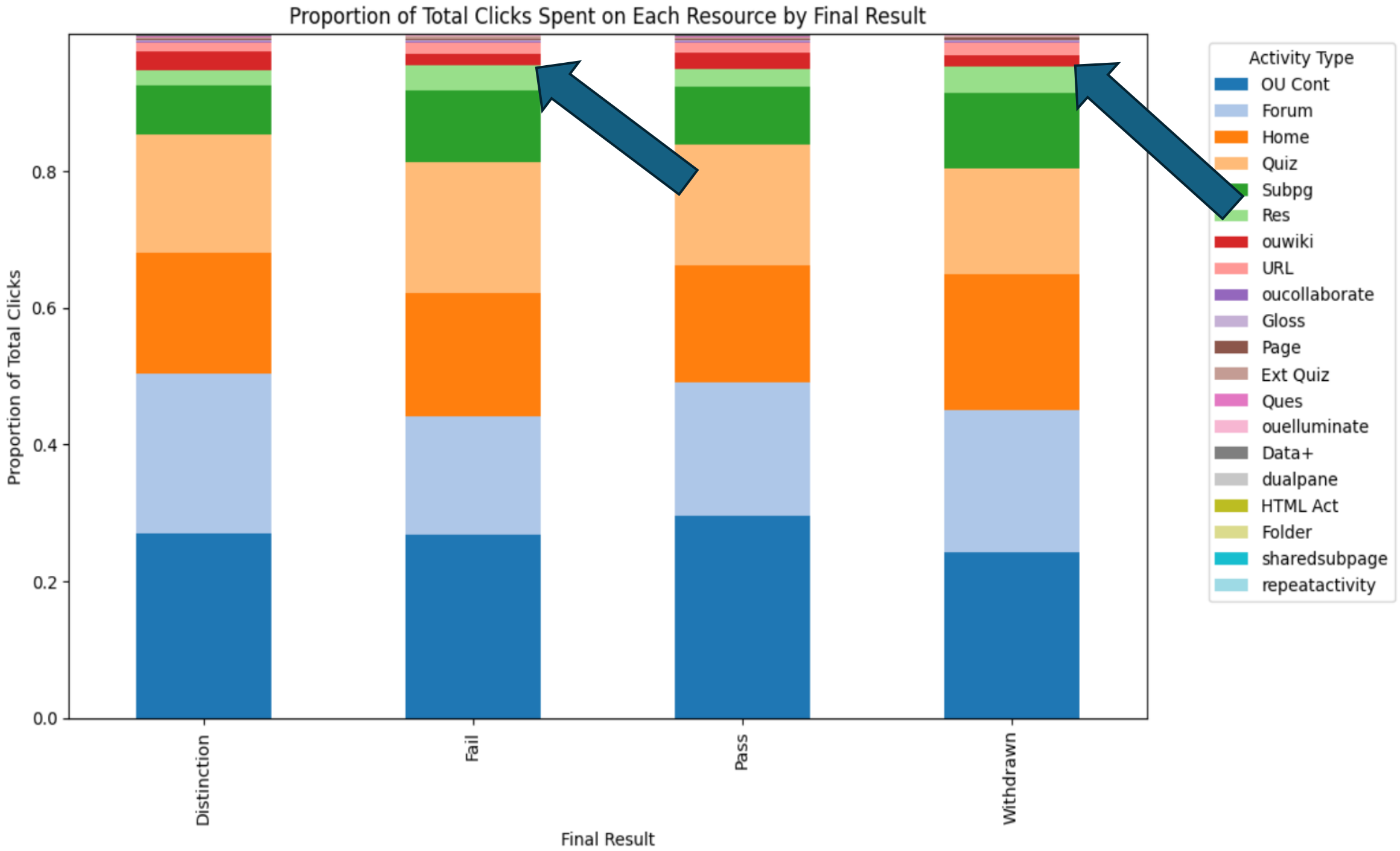
Proportion of Total Clicks Spent on Each Resource by Final Result
Proportion of Total Clicks
0.0
0.2
0.4
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0.8
Distinction
Fail
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Activity Type
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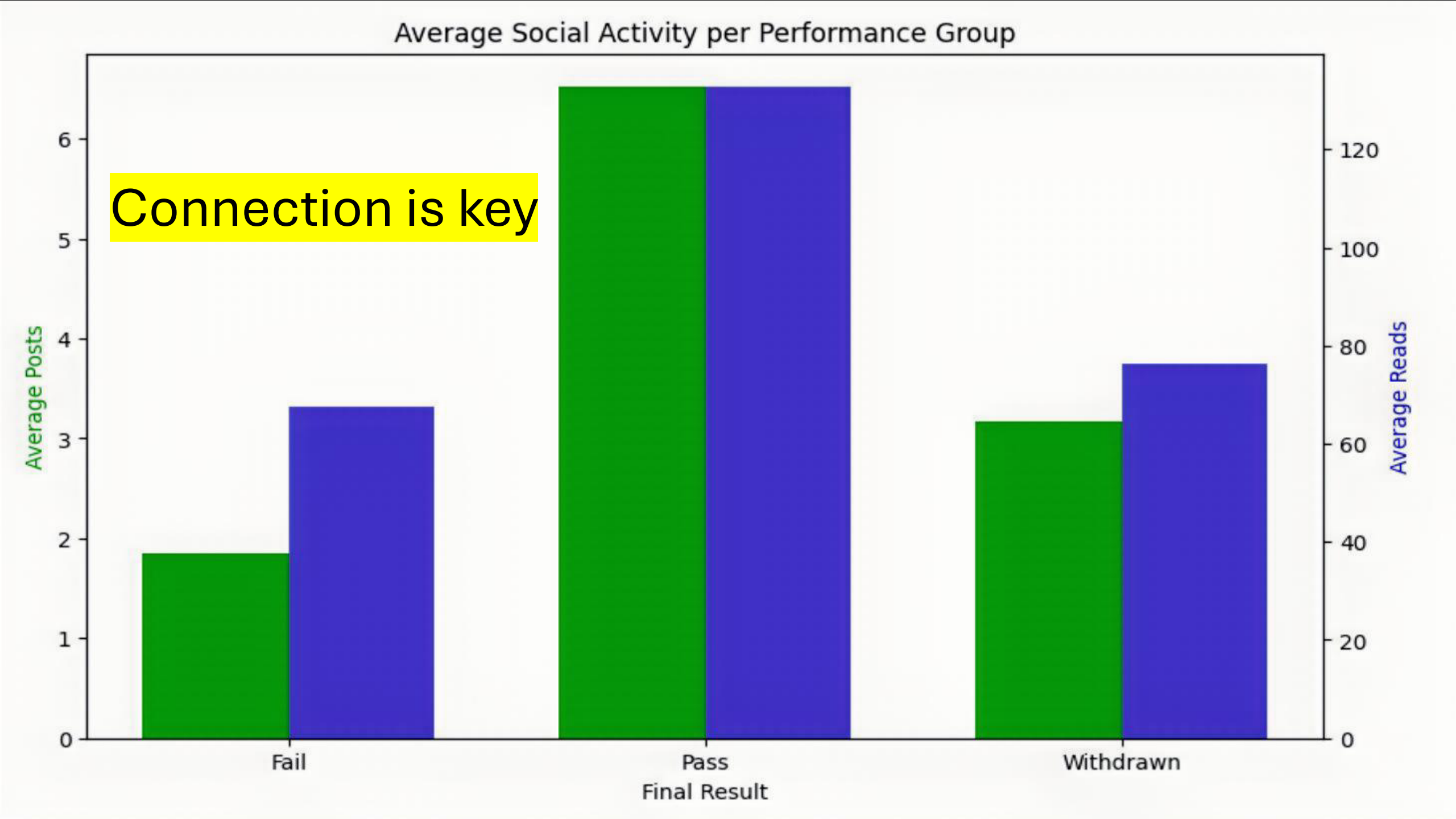
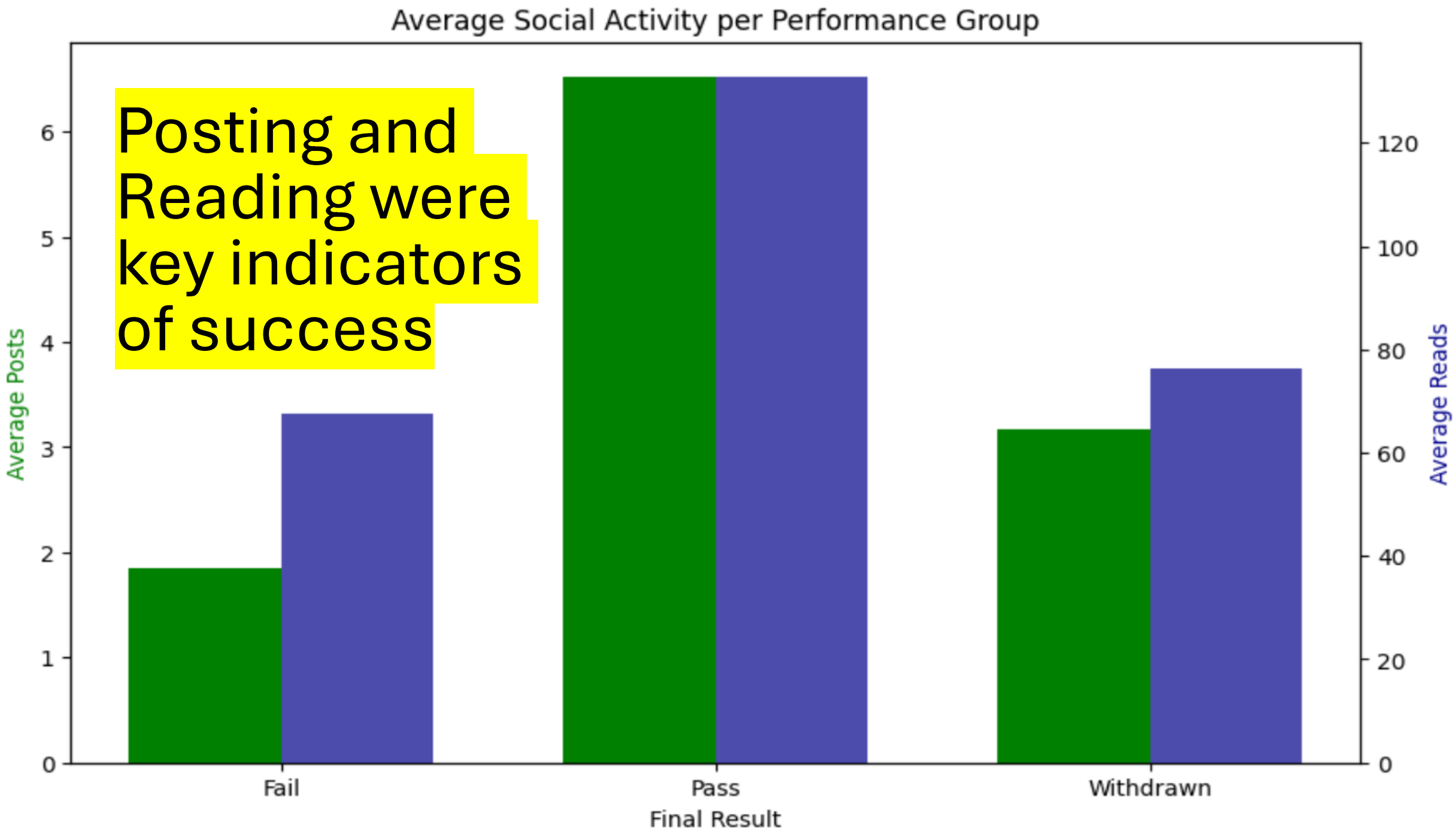
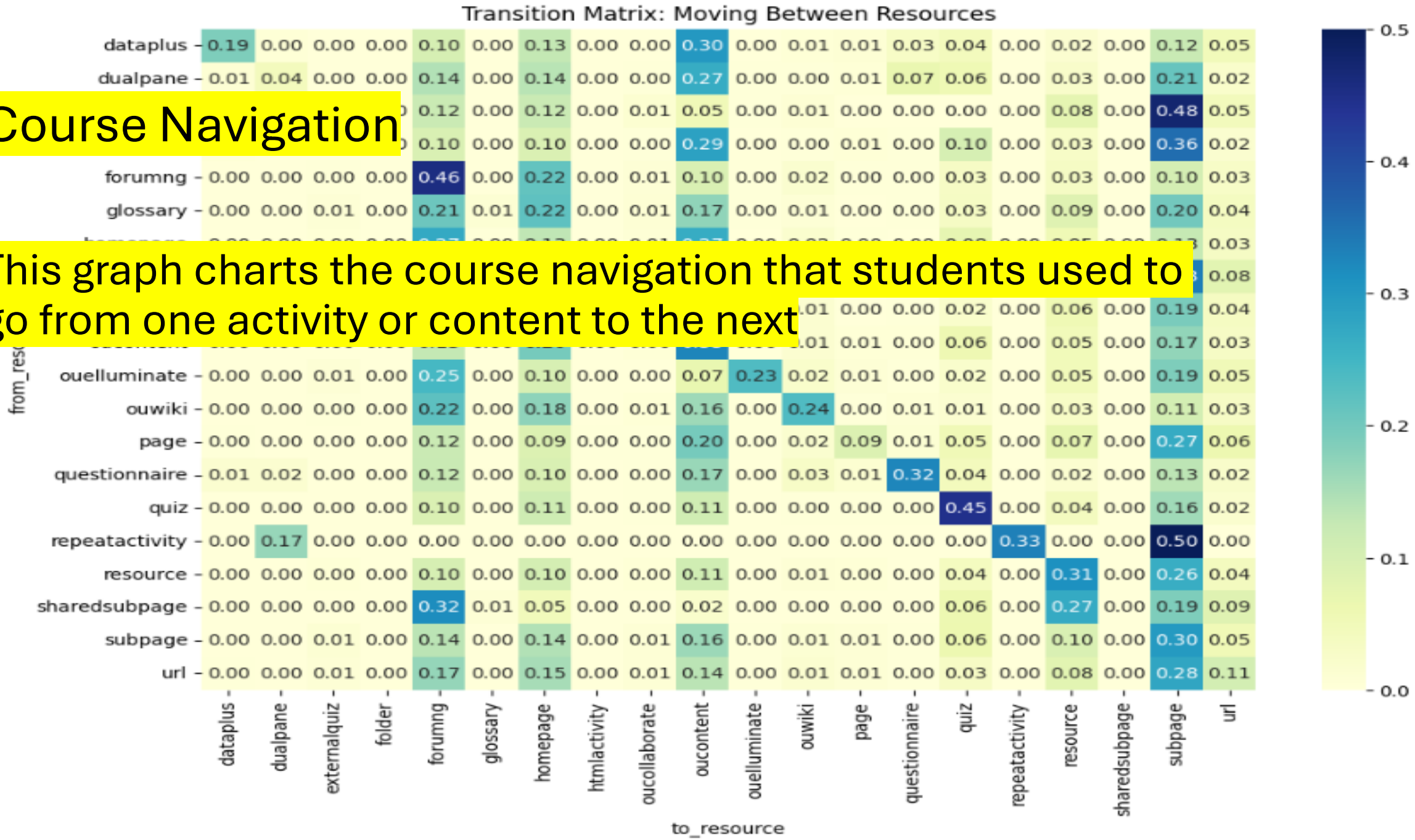
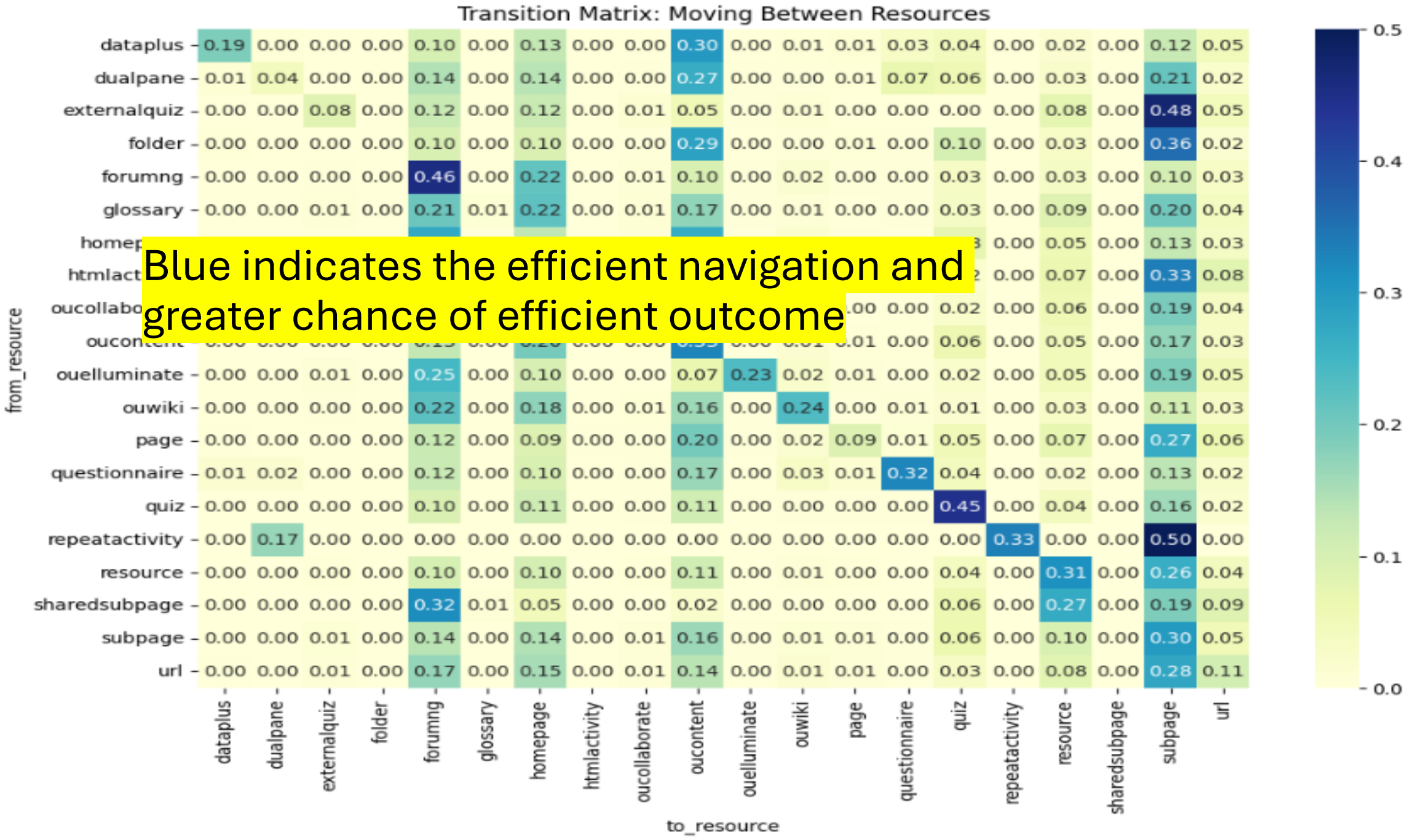

Average Social Activity per Performance Group
Connection is key
Average Posts
Average Reads
0
1
2
3
4
5
6
0
20
40
60
80
100
120
Fail
Pass
Withdrawn
Final Result


Average Social Activity per Performance Group
Posting and Reading were key indicators of success
Average Posts
Average Reads
0
1
2
3
4
5
6
0
20
40
60
80
100
120
Fail
Pass
Withdrawn
Final Result

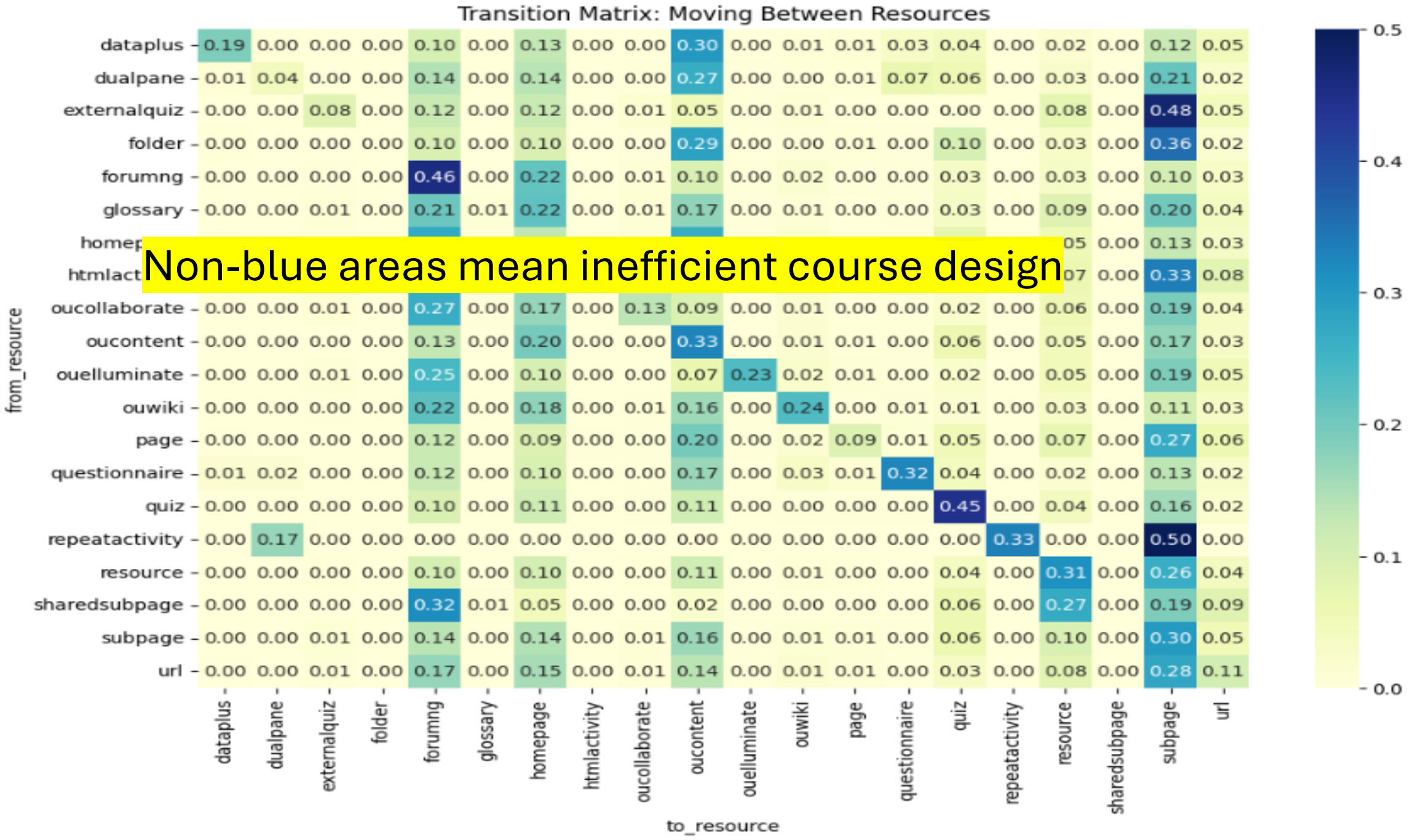
Course Navigation

This graph charts the course navigation that students used to go from one activity or content to the next

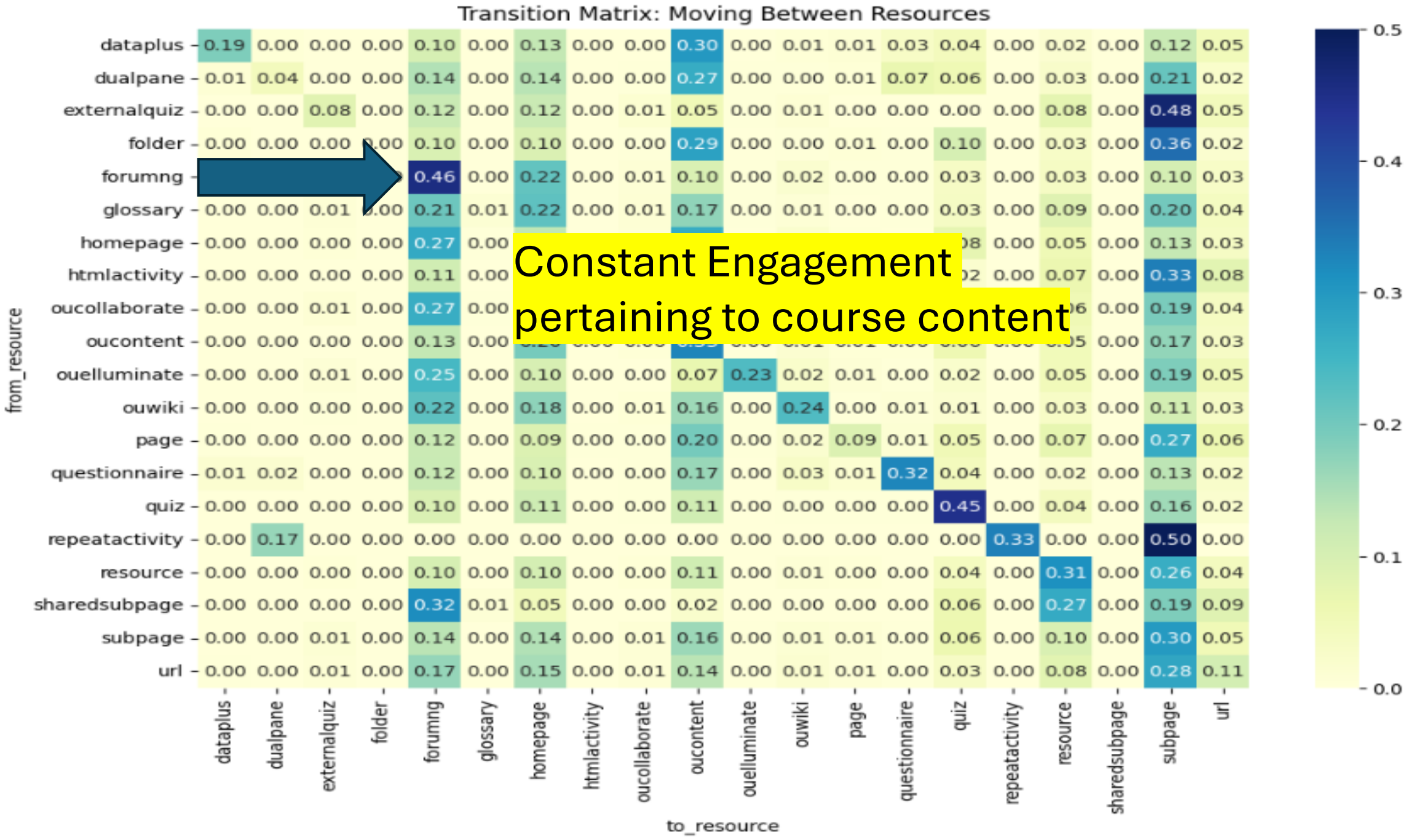


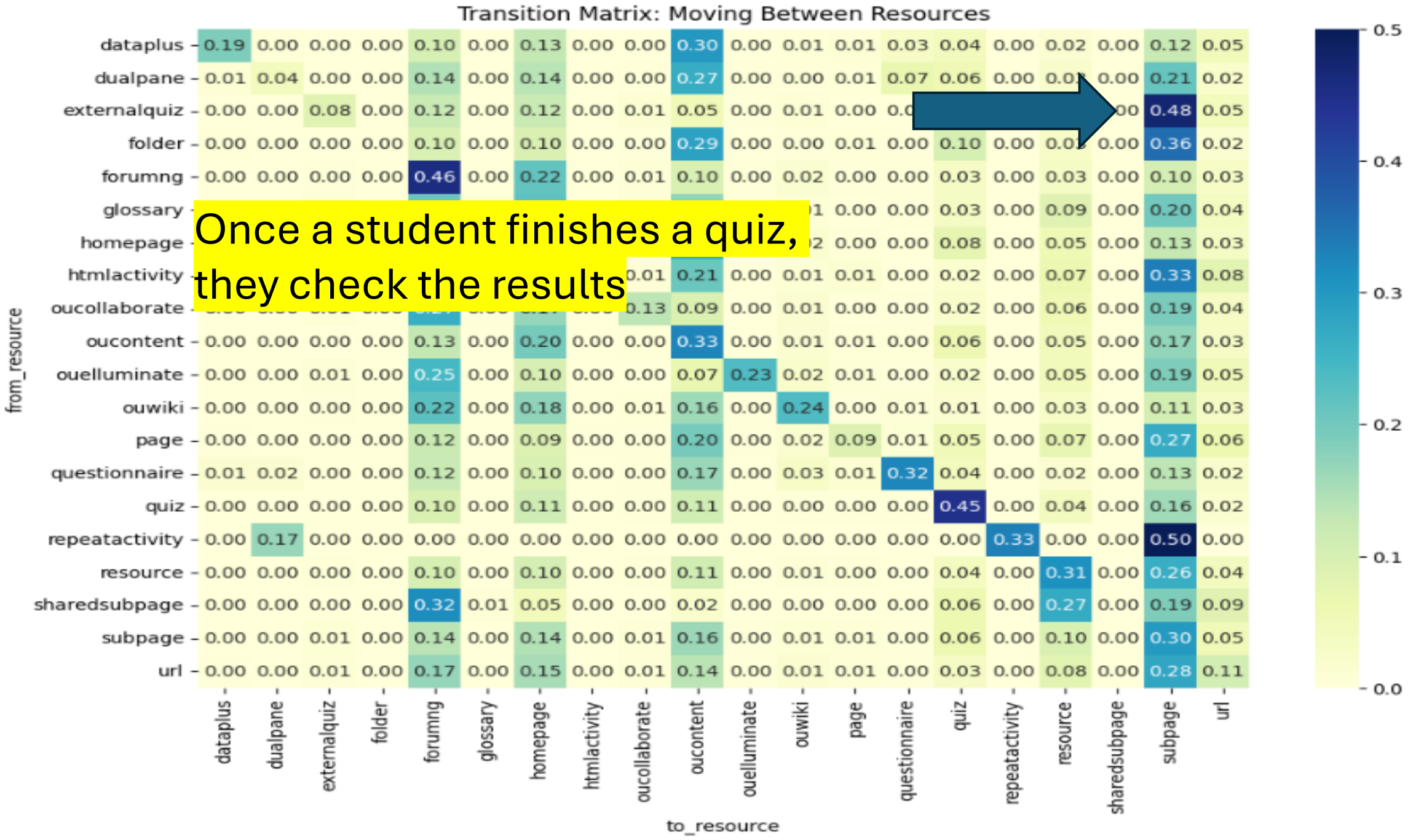


Blue indicates the efficient navigation and greater chance of efficient outcome



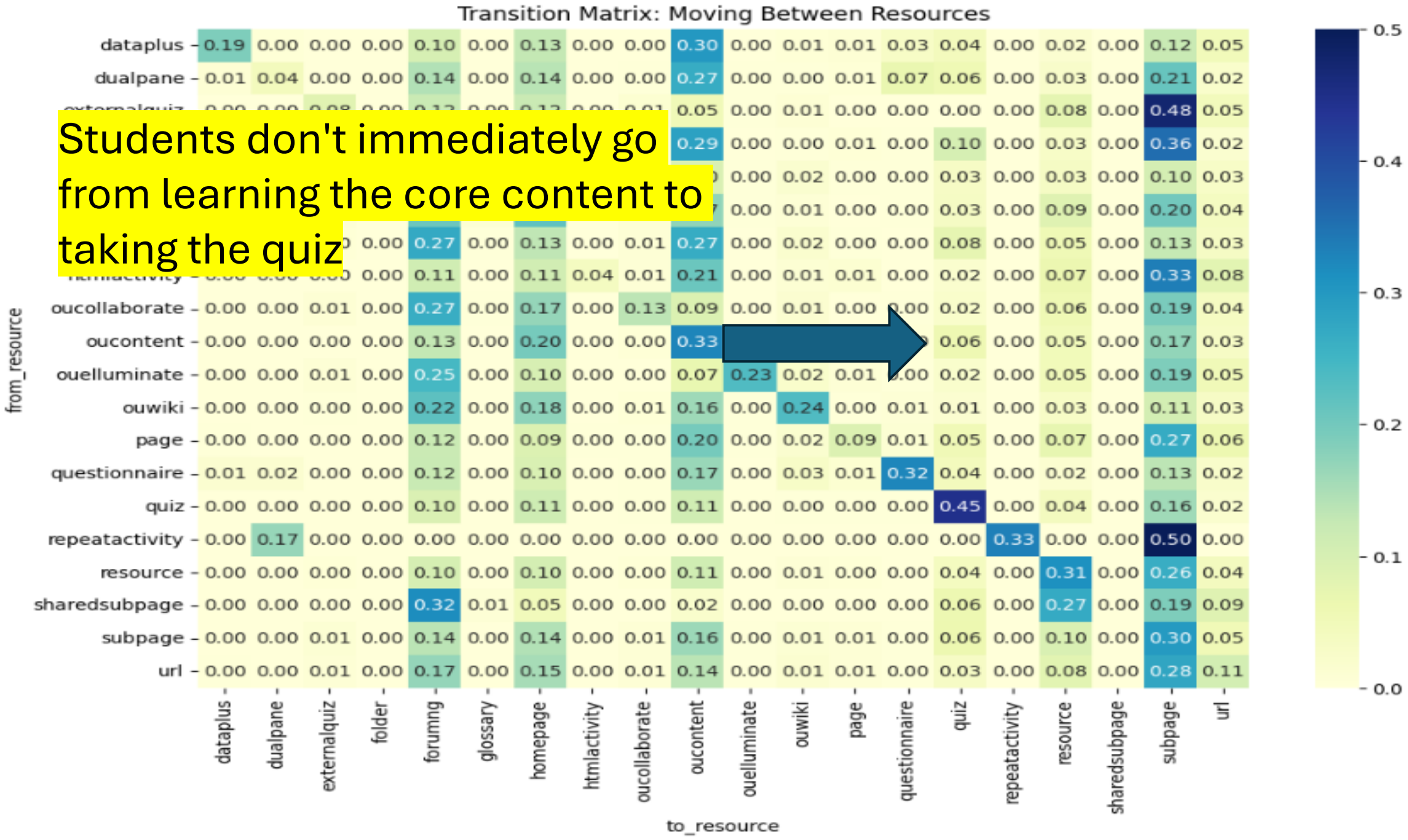
Non-blue areas mean inefficient course design

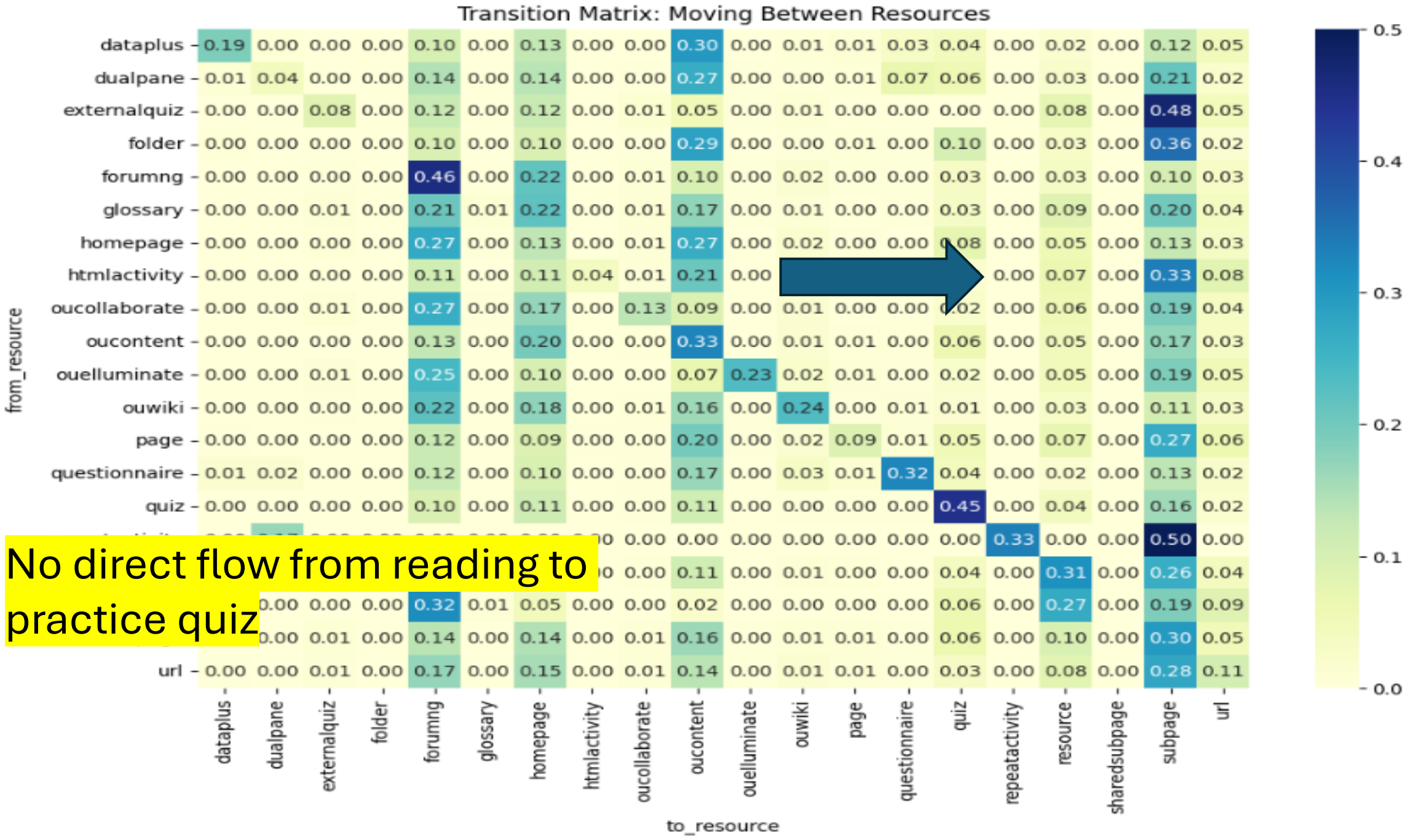

Transition Matrix: Moving Between Resources
from_resource
to_resource
dataplus 0.19 0.00 0.00 0.00 0.10 0.00 0.13 0.00 0.00 0.30 0.00 0.01 0.01 0.03 0.04 0.00 0.02 0.00 0.12 0.05
dualpane 0.01 0.04 0.00 0.00 0.14 0.00 0.14 0.00 0.00 0.27 0.00 0.00 0.01 0.07 0.06 0.00 0.03 0.00 0.21 0.02
externalquiz 0.00 0.00 0.08 0.00 0.12 0.00 0.12 0.00 0.01 0.05 0.00 0.01 0.00 0.00 0.00 0.00 0.08 0.00 0.48 0.05
folder 0.00 0.00 0.00 0.00 0.10 0.00 0.10 0.00 0.00 0.29 0.00 0.00 0.01 0.00 0.10 0.00 0.03 0.00 0.36 0.02
forumng 0.46 0.00 0.22 0.00 0.01 0.10 0.00 0.02 0.00 0.00 0.03 0.00 0.03 0.00 0.10 0.03
glossary 0.00 0.00 0.01 0.00 0.21 0.01 0.22 0.00 0.01 0.17 0.00 0.01 0.00 0.00 0.03 0.00 0.09 0.00 0.20 0.04
homepage 0.00 0.00 0.00 0.00 0.27 0.00 0.05 0.00 0.13 0.03
htmlactivity 0.00 0.00 0.00 0.00 0.11 0.00 0.00 0.07 0.00 0.33 0.08
oucollaborate 0.00 0.00 0.01 0.00 0.27 0.00 0.00 0.19 0.04
oucontent 0.00 0.00 0.00 0.00 0.13 0.00 0.00 0.17 0.03
ouelluminate 0.00 0.00 0.01 0.00 0.25 0.00 0.10 0.00 0.00 0.07 0.23 0.02 0.01 0.00 0.02 0.00 0.05 0.00 0.19 0.05
ouwiki 0.00 0.00 0.00 0.00 0.22 0.00 0.18 0.00 0.01 0.16 0.00 0.24 0.00 0.01 0.01 0.00 0.03 0.00 0.11 0.03
page 0.00 0.00 0.00 0.00 0.12 0.00 0.09 0.00 0.00 0.20 0.00 0.02 0.09 0.01 0.05 0.00 0.07 0.00 0.27 0.06
questionnaire 0.01 0.02 0.00 0.00 0.12 0.00 0.10 0.00 0.00 0.17 0.00 0.03 0.01 0.32 0.04 0.00 0.02 0.00 0.13 0.02
quiz 0.00 0.00 0.00 0.00 0.10 0.00 0.11 0.00 0.00 0.11 0.00 0.00 0.00 0.00 0.45 0.00 0.04 0.00 0.16 0.02
repeatactivity 0.00 0.17 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.33 0.00 0.00 0.50 0.00
resource 0.00 0.00 0.00 0.00 0.10 0.00 0.10 0.00 0.00 0.11 0.00 0.01 0.00 0.00 0.04 0.00 0.31 0.00 0.26 0.04
sharedsubpage 0.00 0.00 0.00 0.00 0.32 0.01 0.05 0.00 0.00 0.02 0.00 0.00 0.00 0.00 0.06 0.00 0.27 0.00 0.19 0.09
subpage 0.00 0.00 0.01 0.00 0.14 0.00 0.14 0.00 0.01 0.16 0.00 0.01 0.01 0.00 0.06 0.00 0.10 0.00 0.30 0.05
url 0.00 0.00 0.01 0.00 0.17 0.00 0.15 0.00 0.01 0.14 0.00 0.01 0.01 0.00 0.03 0.00 0.08 0.00 0.28 0.11
dataplus dualpane externalquiz folder forumng glossary homepage htmlactivity oucollaborate oucontent ouelluminate ouwiki page questionnaire quiz repeatactivity resource sharedsubpage subpage url
0.5
0.4
0.3
0.2
0.1
0.0
Constant Engagement pertaining to course content



Once a student finishes a quiz, they check the results

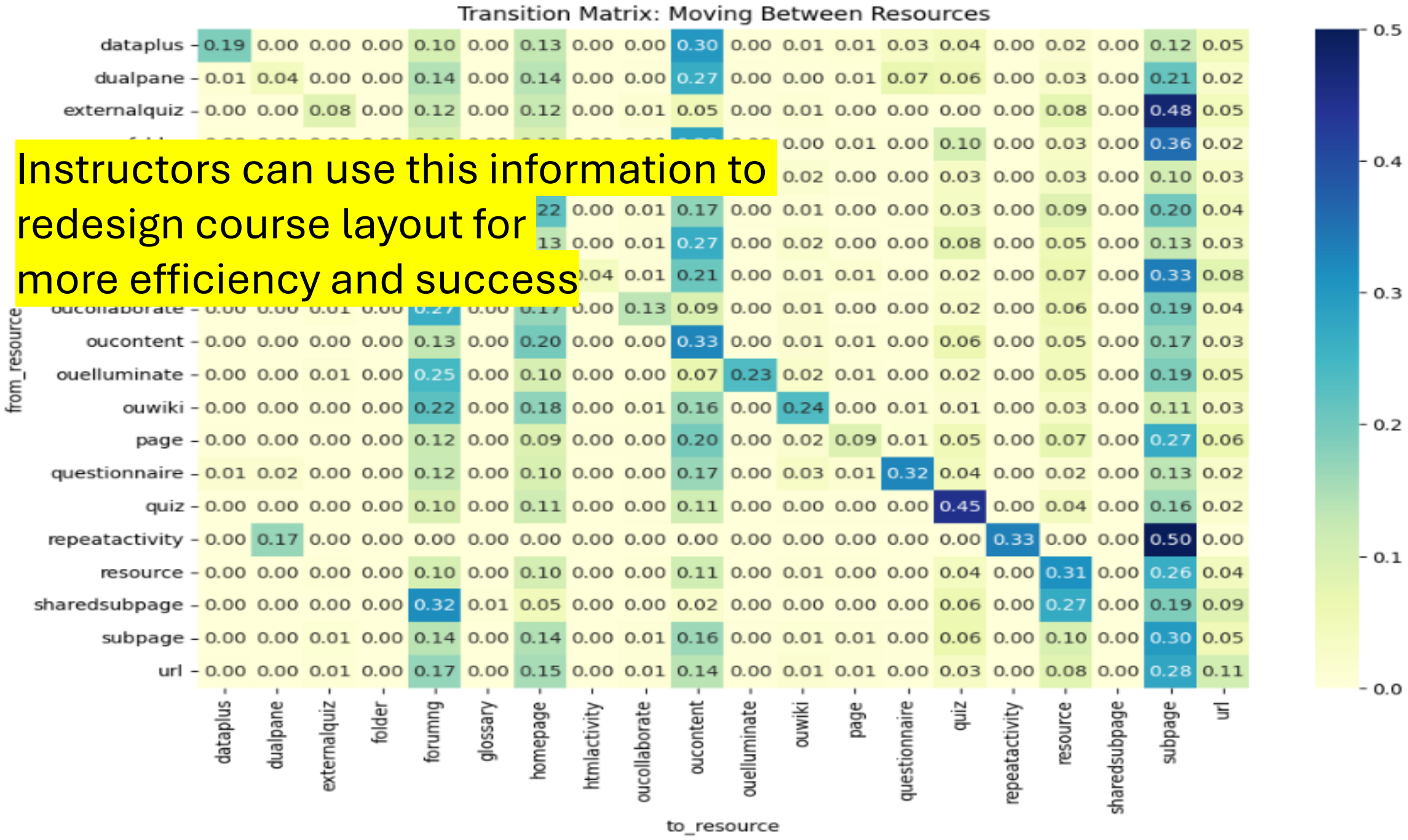
Students don't immediately go from learning the core content to taking the quiz





No direct flow from reading to practice quiz

Instructors can use this information to redesign course layout for more efficiency and success



From Analysis to Prediction

- Use early-course data from the first 20% of the course
- Predict student outcomes before the course ends
- Enable early identification of at-risk students

Machine Learning Approach

Using early-course features to predict student outcomes

Prediction Tasks

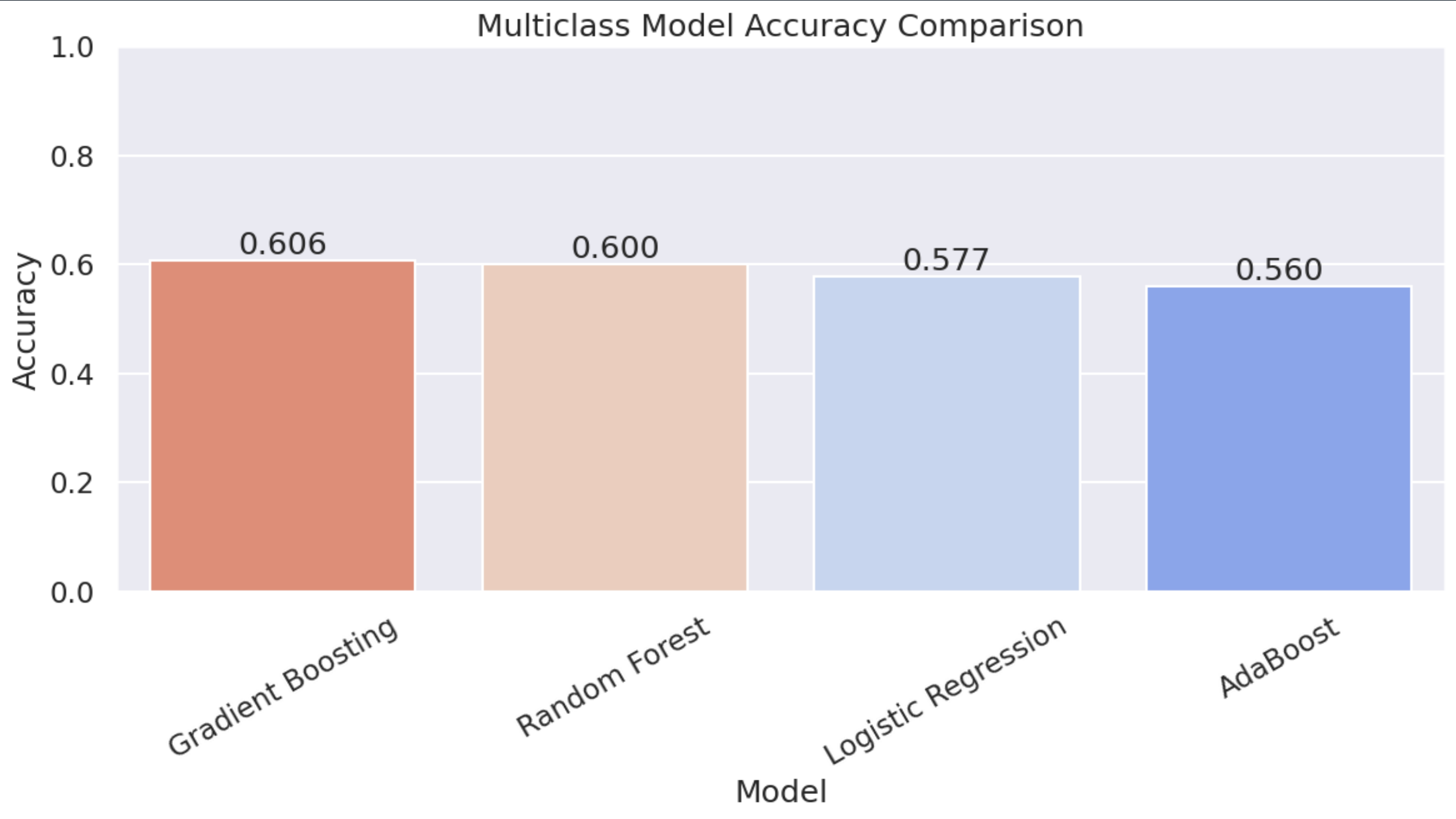
- Multiclass classification
  - Distinction
  - Pass
  - Fail
  - Withdrawn
- Binary classification
  - At-risk
  - Not at-risk

Features Used for Prediction

- Early engagement
  - Total clicks
  - Active days
- Early assessment performance
  - Average score
  - Max score
- Student information
  - Education
  - Age
  - Region
- Registration behavior

Models Tested

- Logistic Regression
- Random Forest
- Gradient Boosting
- AdaBoost

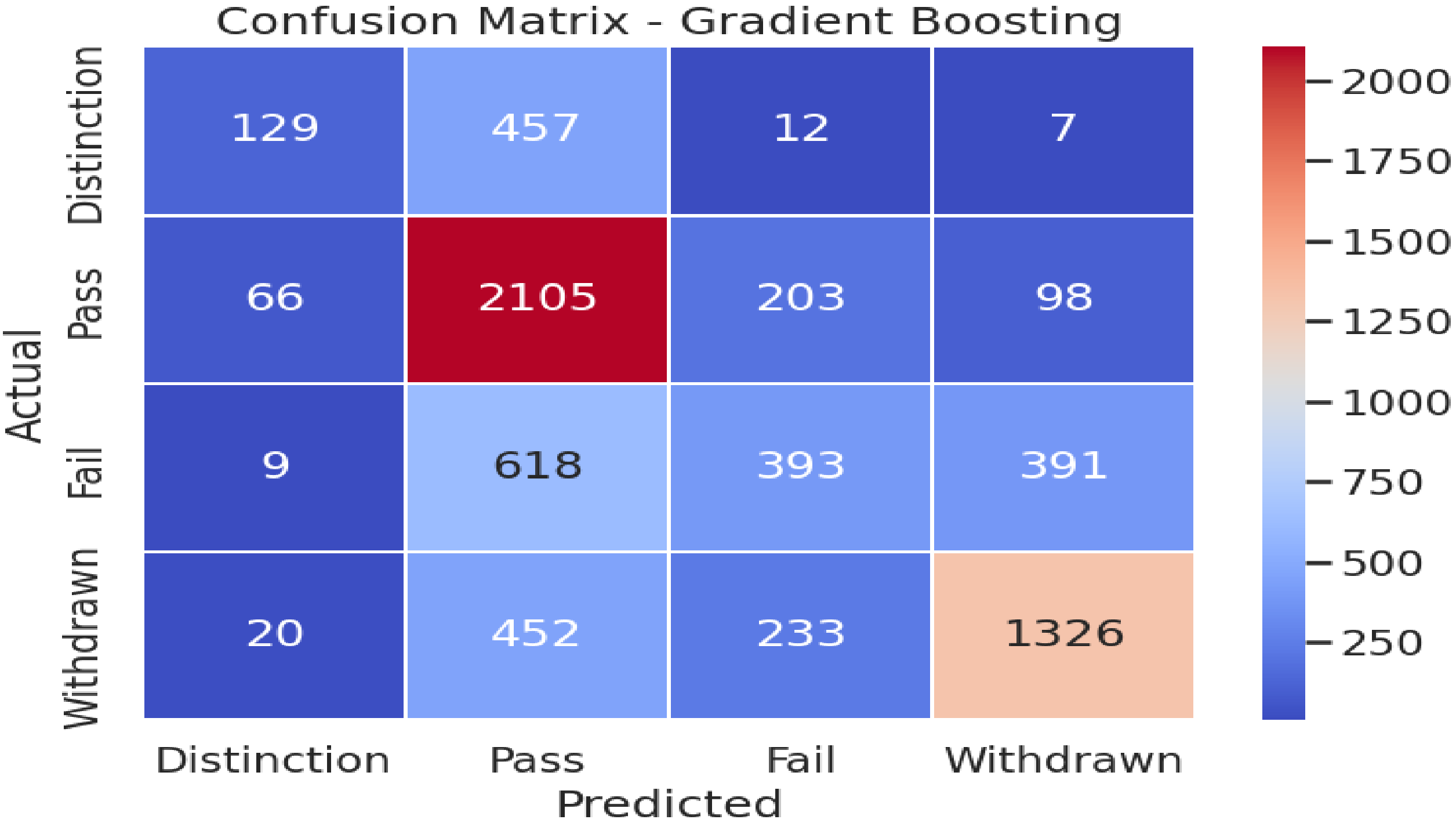

Multiclass Model Accuracy Comparison
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0.606
0.600
0.577
0.560
Gradient Boosting
Random Forest
Logistic Regression
AdaBoost
Model

Multiclass Prediction Results

- Best model: Gradient Boosting
- Accuracy: 60.6%
- Hardest classes:
  - Distinction
  - Fail
- Strongest predictions:
  - Pass
  - Withdrawn

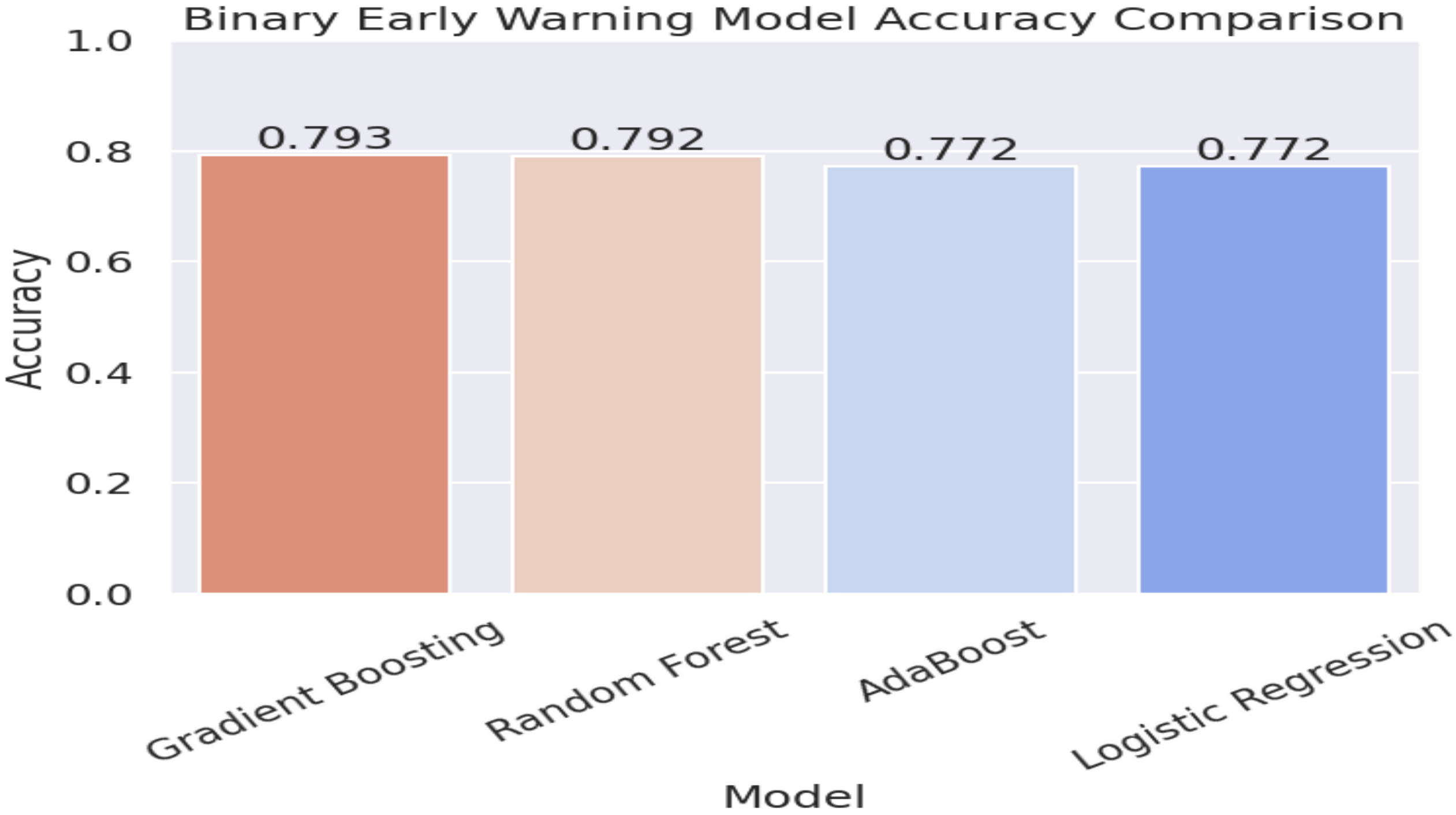
Multiclass Confusion Matrix

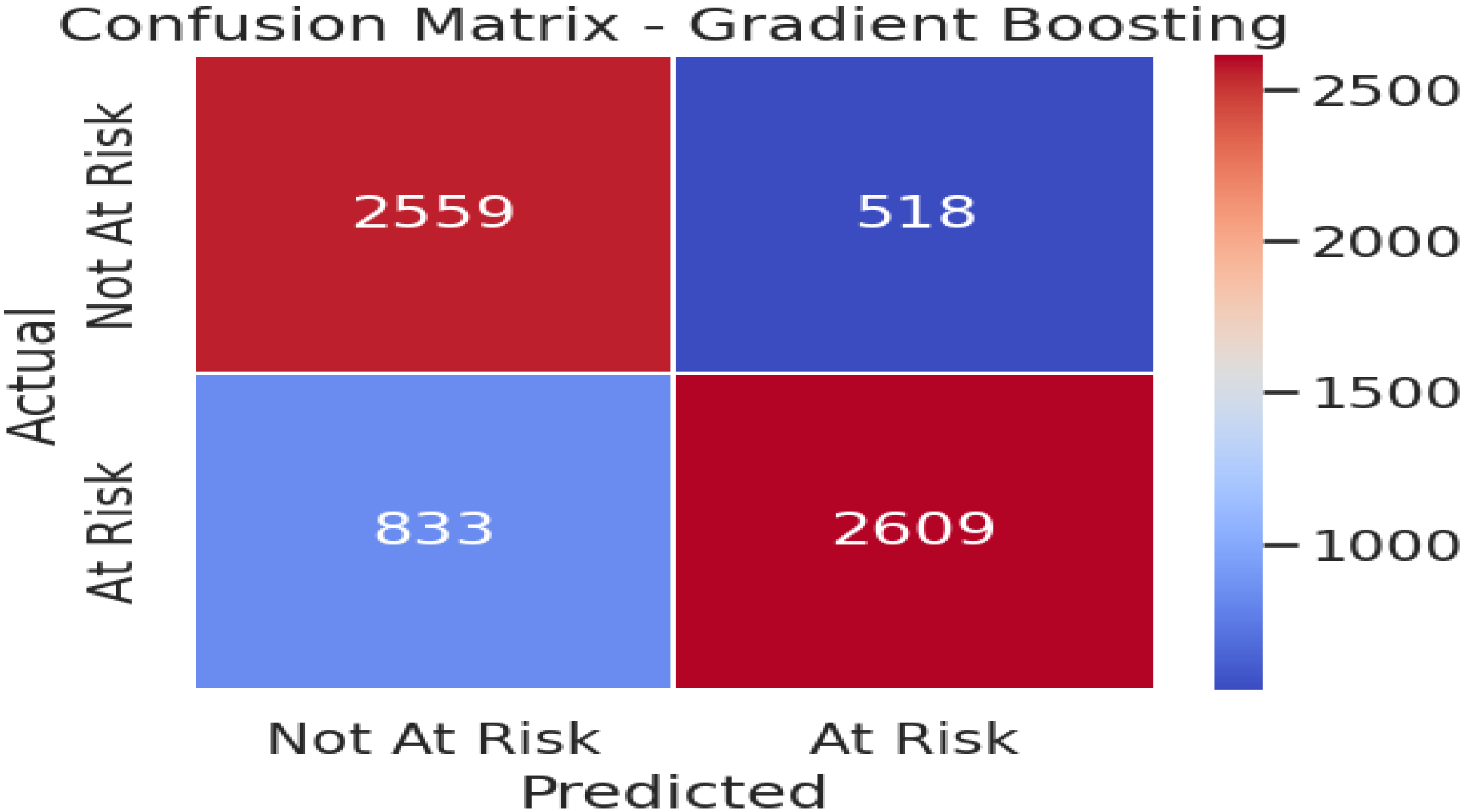
- Best performance:
  - Pass
  - Withdrawn
- More confusion between:
  - Distinction
  - Fail


Confusion Matrix - Gradient Boosting
Actual
Distinction
Pass
Fail
Withdrawn
129
457
12
7
66
2105
203
98
9
618
393
391
20
452
233
1326
Distinction
Pass
Fail
Withdrawn
Predicted
2000
1750
1500
1250
1000
750
500
250

Early Warning Prediction

- Binary classification
- Best model: Gradient Boosting
- Accuracy: 79.3%
- Designed for early intervention

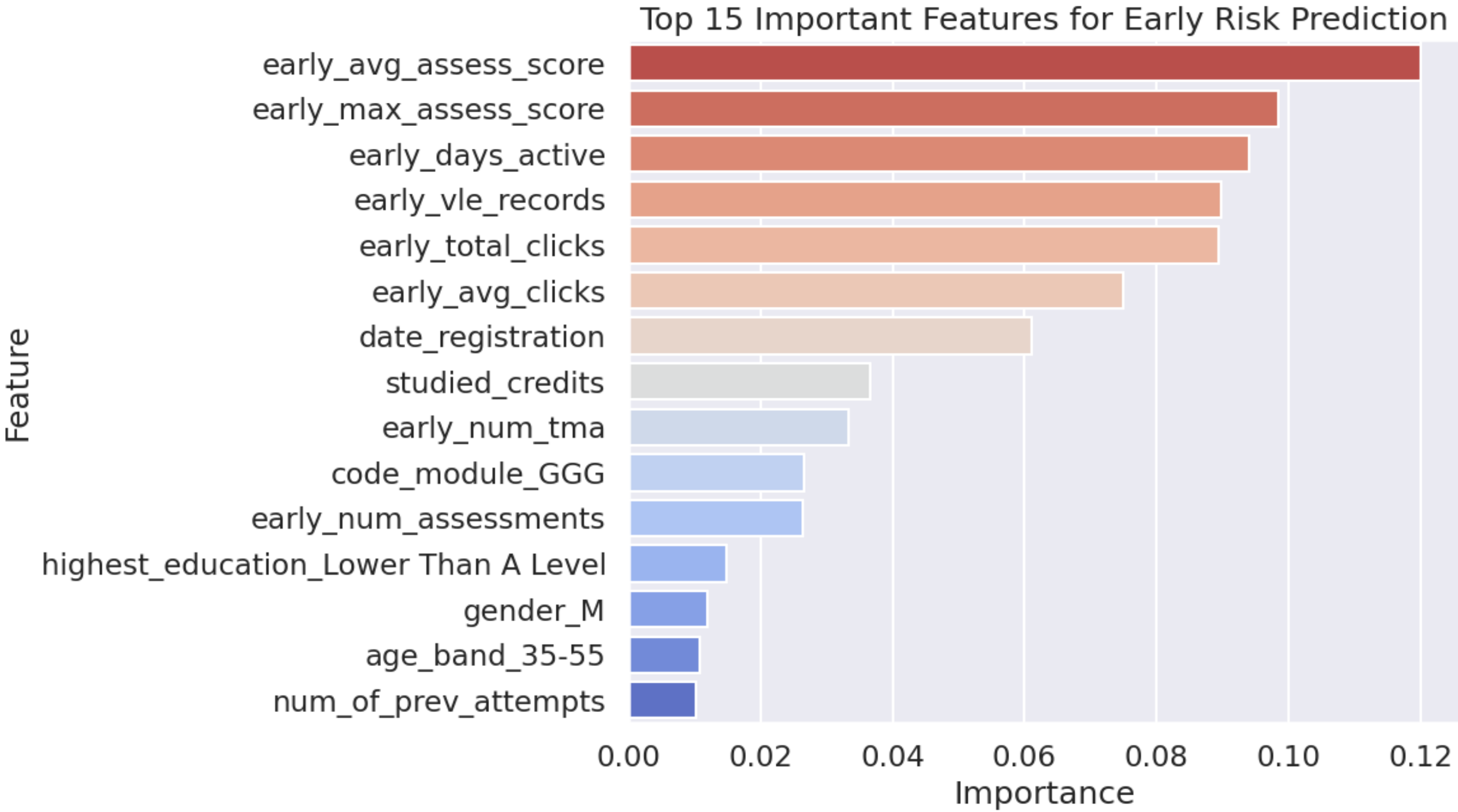

Binary Early Warning Model Accuracy Comparison
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
0.793
0.792
0.772
0.772
Gradient Boosting
Random Forest
AdaBoost
Logistic Regression
Model


Confusion Matrix - Gradient Boosting
Actual
Not At Risk
At Risk
2559
518
833
2609
Not At Risk
At Risk
Predicted
2500
2000
1500
1000

Feature Importance

Top predictors:

- Early average assessment score
- Early maximum assessment score
- Early active days
- Early VLE records
- Early total clicks


Top 15 Important Features for Early Risk Prediction
early_avg_assess_score
early_max_assess_score
early_days_active
early_vle_records
early_total_clicks
early_avg_clicks
date_registration
studied_credits
early_num_tma
code_module_GGG
early_num_assessments
highest_education_Lower Than A Level
gender_M
age_band_35-55
num_of_prev_attempts
Feature
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance

Causal Effects: From Correlation to Causation

Correlation: Students who click more tend to succeed

Limitation: Correlation does not imply causation

Key Question:

- Does engagement *cause* success?
- What happens if we increase engagement?

- Approach:
  - Compare similar students
  - Control for background factors

Causal Effect of Engagement on Success

Treatment: High engagement (high clicks)

Outcome: Passing the course

Result (Average Treatment Effect (ATE)):

- **+32.3% increase in pass probability**

Interpretation:

- Increasing engagement improves success

Implication:

- Early intervention strategies can be effective

Final Machine Learning Insight

- Early behavior predicts later outcomes
- Engagement and assessment matter most
- Early warning is realistic and useful
- Machine learning can support intervention

Sources

Kuzilek, J., Hlosta, M. & Zdrahal, Z. Open University Learning Analytics dataset. *Sci Data* **4**, 170171 (2017). https://doi.org/10.1038/sdata.2017.171

The Open University. (n.d.). *Facts and figures*. https://www.open.ac.uk/policies-and-reports/facts-and-figures *(Accessed April 13, 2026)*

Sim, S. (2024). *Analysis of the OULAD dataset*. GitHub Pages. https://sergiosim.github.io/OULAD/

Thank you!

Questions?